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GDMS Oracle TLS Configuration Guide (12cR1+)

SECURITY WARNING

Contents

Purpose

The purpose of this standard document is to provide an authoritative single-source reference summarizing the steps to configuring a database to support TLS (SSL) sqlnet traffic. This is required for any databases containing restricted, highly restricted or PCI/KFS/banking data, which should be encrypted in transport.

The contents of this document supersede all documents published prior.

Document Revision History

The revision history shows the history for this document and provides descriptions of particular changes made.

Version	Date	Description	Name
1.0	22-May-2017	First Version	Alan Goodall
1.1	6-Jul-2017	Additional details and instructions added to all sections, and new sections for EM13c and mandatory patches for GoldenGate for TLS.	Alan Goodall
1.2	12-Dec-2017	Updated following feedback from MegaGrid team.	Alan Goodall
1.3	10-Apr-2018	Added warning to ensure different wallets are used for TLS and TDE.	Alan Goodall
1.4	15-May-2018	Corrected a mistake in local_listener parameter, and added more examples of setting the parameter correctly, and ensuring sqlnet.ora is present in the DB home. Added a warning that local_listener must be set on all 11.2 databases on a 12.1 cluster.	Alan Goodall
1.5	16-May-2018	Added additional steps to ensure sqlnet.ora is correctly set in a RAC environment, and further examples of testing connections.	Alan Goodall
1.6	10-Oct-2018	Changed location of wallet to /u01/app/oracle/admin/tls_wallet, so that it is preserved during GI rolling upgrade.	Alan Goodall
1.7	17-Dec-2018	Added additional connectivity test using openssl.	Alan Goodall
1.8	19-Mar-2019	Updated Weblogic section to include additional parameter to ensure it works after Database January 2019 PSU.	Alan Goodall

Version	Date	Description	Name
1.9	20-May-2019	Added link to pre-prepared Oracle client wallets. Added additional information for requesting a certificate, following new process. Updated references to new CA “Dell Technologies Root Certificate Authority 2018”	Alan Goodall
2.0	9-Jul-2019	Amended instructions to use new automation tool to configure server. Added Appendix D on troubleshooting different errors	Alan Goodall
2.0.1	5-Sep-2019	Minor change to command to configure GRID for TLS, and inclusion of rolling listener bounce.	Alan Goodall
2.0.2	30-Sep-2019	Clarification of the automation steps following feedback.	Alan Goodall
2.1.0	20-Dec-2019	Added instructions for configuring SQL developer to use TCPS/TLS connections. Added section on OUD naming standards for TLS.	Alan Goodall
2.2.0	24-Mar-2020	Added section covering process to check expiry dates and email contact information for a certificate.	Alan Goodall
2.2.1	21-Apr-2020	Added help for error ORA-28862 which indicates a firewall issue.	Alan Goodall
2.3.0	18-May-2020	Added application server configuration instructions for ODP.Net, and Java (non Weblogic).	Alan Goodall
2.3.1	19-May-2020	Added SSL_CIPHER_SUITES parameter to sqlnet.ora and listener.ora to force Dell approved cipher suites.	Alan Goodall
2.4.0	21-May-2020	Added new section on “Removing TCP/1521 Configuration”	Alan Goodall

Version	Date	Description	Name
2.4.1	22-May-2020	Amended the section “Removing TCP/1521 Configuration” to reference needing to correct ASM local_listener parameter.	Alan Goodall
2.4.2	28-May-2020	Added warning about SSL_CIPHER_SUITES and Oracle 12.1 clients, as they do not support the Dell approved cipher suites.	Alan Goodall
2.4.3	12-Jun-2020	Added TLS 1.2 compatibility matrix.	Alan Goodall
2.5.0	16-Jun-2020	Added requirement that restricted, highly restricted or PCI/KFS/banking data must be encrypted in transport using TLS 1.2.	Alan Goodall
2.5.1	17-Jun-2020	Added Quick TLS Configuration section to reference the default wallet created on all new builds, with certificate CN=oracledb.dev.amer.dell.com.	Alan Goodall
2.6.0	03-Jul-2020	Added information for how to TLS enable L-EMC servers. Updated Appendix D to include troubleshooting ‘Segmentation fault’ error.	Alan Goodall

Approval History

The approval history shows the history for this document.

Date	Description	Name
	Infrastructure Management Engineering Manager	Ravi Kulkarni

Overview

Traffic between any Oracle Client and an Oracle database uses Oracle SQLNET, which typically runs over TCP. This is an Oracle proprietary protocol which out the box doesn't provide any type of encryption of the traffic (except for the handshake providing username/password which is encrypted).

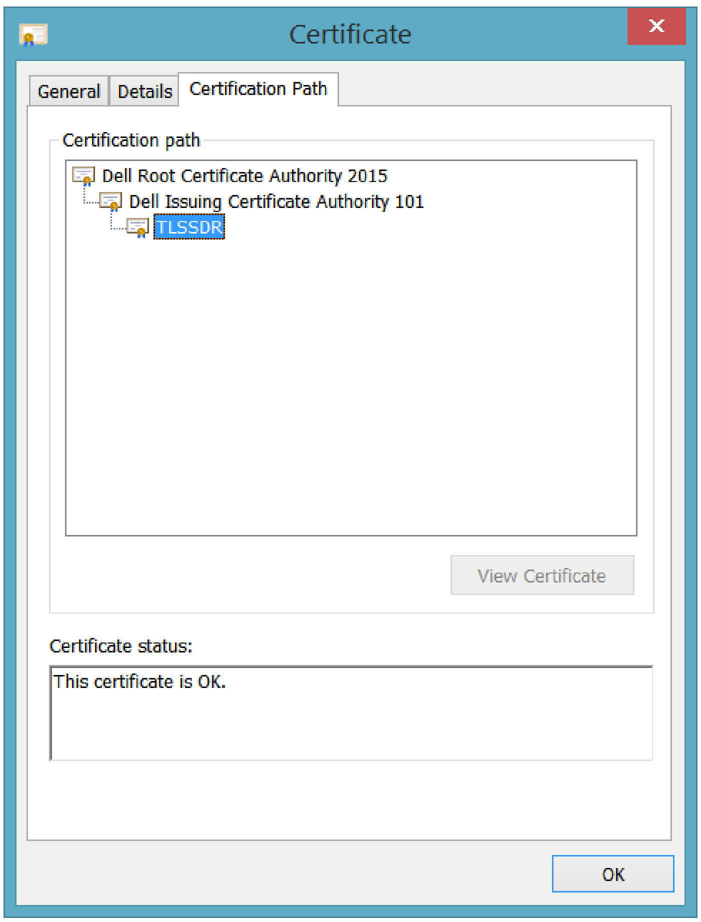
For databases containing restricted, highly restricted or PCI/KFS/banking data, the Dell standard is all network traffic must be encrypted using transport layer security (TLS). In the past this recommendation just covered the traffic from a client (e.g. HTTPS connections from a browser) to the application server but has been extended to cover the traffic from application servers or other clients to the database.

This document describes the process to configure a database server to encrypt sqlnet traffic using TLS. It also includes examples of how different clients might be configured to connect using TLS.

Background

To help understand the process of configuring TLS for a database, it is worth defining some concepts first:

1. TLS stands for transport layer security, and is now an industry standard mechanism for encrypting network traffic. It is an evolution of secure sockets layer (SSL), which is now considered obsolete, and vulnerable to network attacks. TLS v1.0 was the first version, and was very similar to SSL v3.0. Both are considered insecure, and it is recommended that TLS v1.1 or 1.2 is used.
2. TLS requires a trusted certificate to encrypt the network traffic. A certificate is issued by a certificate authority, and when you view the certificate, it will have a hierarchy of certificates from the server certificate (at the lowest level) all the way up to the root certificate (which is usually the trusted certificate for the certificate authority). When a server certificate is opened in Windows, you can view the certification path:



In the example above, the 'Dell Root Certificate Authority 2015' is the root certificate and has been issued by the Dell Certificate Authority (Dell CA for short).

3. The steps for requesting a certificate are:
    a. Create a database wallet.
    b. Create a new private key which will be used to encrypt all traffic.
    c. Create a certificate signing request (CSR).
    d. Submit the CSR to a trusted certificate authority, and wait for them to issue a public key certificate.

   e. Add the certificate to the database wallet, ensuring that the other trusted certificates in the certificate chain are also added to the database wallet.

4. For a client to access a server which is using TLS, it must be able to trust the server certificate. In order to do that, it must have a keystore in which to check whether the certificate is valid. If it can't find the certificate in the keystore, it will walk up the chain to the top level certificate (root CA), and as long as that root CA is in the keystore, it will then trust the server certificate and perform the TLS (SSL) handshake. If it can't find the root certificate in the keystore it will reject the connection.
5. Different platforms and software have different trusted keystores:
   a. Windows has its own keystore (called Microsoft Certificate Store/MCS), which Microsoft periodically updates with new trusted CA's. When a browser is installed on Windows, it automatically uses the Windows keystore to check whether a certificate is valid. It does not contain the Dell CA, but the Dell standard laptop/server build should add the Dell CA. The certmgr.msc tool is used in Windows to manage the certificate store.
   b. Linux has its own keystore. As Linux uses openssl to manage certificates, the default location for the keystore is /etc/ssl/certs. All new builds include the Dell CA, but older builds will miss this.
   c. Java has its own keystore, and doesn't have any mechanism to devolve certificate checking to the OS. This means that if the root certificate doesn't exist in the Java keystore, Java will fail to initiate a TLS connection to a server. Java is shipped with a trusted keystore which contains many trusted certificate authorities. It does not contain the Dell CA. A Java wallet is based on propriety java format called JKS.
   d. Oracle has its own keystore, and by default is not shipped with any keystore. You have to create a wallet and import any trusted certificates manually into the wallet. An Oracle wallet is based on the PKCS#12 standard. PKCS#12 wallets usually have an extension .p12.
6. There are different certificate standards, and different tools require different formats:
   a. X.509 is a standard that defines the format of public key certificates.
   b. PEM (Privacy-enhanced Electronic Mail) is an ASCII based file format for storing X.509 certificates. Commonly used file extensions are .pem, but often .cer and .crt will be used. These certificates when viewed in an editor will have a '-----BEGIN CERTIFICATE-----' at the start of the file, and an '-----END CERTIFICATE-----' at the end of the file. The actual certificate is the text between these.
   c. DER is a binary version of a PEM certificate. It can only be read using a tool like openssl. These often also use an extension .cer, .crt or .der.
   d. PKCS#7 is a format that contains one or more certificates. Usually the file extension will be .p7b.
   e. PKCS #10 is a format for certificate signing requests (CSR's). This can be a binary or ASCII format file. If base-64 encoded ASCII format, the file will contain the certificate request, between the '-----BEGIN CERTIFICATE REQUEST-----' and '-----END CERTIFICATE REQUEST-----' tags.
   f. The Dell certificate authority usually issues a certificate in PEM format, with a .cer extension.

The following Metalink Note is the master note for all Oracle TLS configuration, and has an overview of the TLS handshake:

[Master Note For SSL/TLS (Doc ID 2229775.1)]

In terms of configuring an Oracle database to use TLS, the following Metalink Note gives details of the process:

[Step by Step Guide: How to Configure SSL / TCPS on ORACLE RAC (with SCAN) (Doc ID 1448841.1)]

That document omits some vital steps which are required if both TLS and non-TLS conncetions are required. The following document has additional detail on configuring the remote_listener parameter on a RAC cluster:

[How To Configure Scan Listeners With A TCPS Port? (Doc ID 1092753.1)]

Finally, for troubleshooting TLS connection issues, the following note is useful:

[SSL Troubleshooting Guide (Doc ID 166492.1)]

The remainder of this document walks through how this was implemented on a Dell DB engineering environment.

N.B. If the database is using TDE, then make sure a new wallet is created for TLS and kept separate from the TDE wallet.

N.B.2. The TDE wallet is usually under /u01/app/oracle/admin/<dbname>/wallet i.e. outside the GRID_HOME. In order to centralise all wallets under here, the TLS wallet location has been moved to /u01/app/oracle/admin/tls_wallet, so that it is preserved during a GI or DB upgrade.

Hardware, Network & OS Requirements

This procedure assumes that the cluster or VM was installed using Dell's automation, or as described in the following documents:

GDMS 12.1.0.2c Cluster DB Deployment

The version of the database must be at least 12.1.0.2, as the Dell CA issued certificates are of a format which is only compatible with Oracle 12.1 and above. This restriction also applies to the client, if the standard Oracle Client is used; it must be at least an Oracle 12.1 client (note that the 12.1 client lacks support for the Dell approved cipher suites so ideally 12.2 client or higher would be used).

Network/IP Address Requirements

There are no specific networking requirements. The TLS configuration will work against whatever listener the database registers with i.e. the VIP listener or the SCAN listener.

Grid Infrastructure Home Directory

This guide assumes that the Grid Infrastructure Home is already installed, and that ASM is configured and running.

The standard installation directory for GRID is /u01/app/grid/product/12.1.0.2/grid_1, and this guide assumes that $GRID_HOME is set to this path.

OS Path

This guide assumes that the unix PATH environment variable has the location of the Dell database tools on the path i.e. /misc/software/oracle/tools/maintenance, as it relies on the tools in that directory.

MegaGrid Requirements

This guide can be used to configure TLS on a MegaGrid environment. There's no real difference between configuring a MegaGrid vs a normal RAC cluster, but note the following:

- If there are 11.2 databases running on the 12.1+ cluster, please ensure the local_listener parameter is set and points only to TCP/port 1521. If the parameter is empty, the database will attempt to register with TCPS/port 1523 and crash on startup.
- A single server certificate should be used for all databases on the MegaGrid.

L-EMC Servers

This guide can be used to configure TLS on L-EMC servers, with the following caveats:

- The dell_orapki_auto.sh tool can be used to generate a CSR and then import the issued Dell certificate.
- The "dell_oracle_tls_auto.sh" tool cannot be used to automate the TLS configuration, but it can be used to validate the configuration once it has been manually configured, using the following command:

```
dell_oracle_tls_auto.sh -status
```

Process Summary

These are the main steps required to configure TLS for a database server/cluster:

1. Create a wallet/certificate signing request and submit to SNOW to receive a new certificate for the server/servers.
2. Import the certificate into the wallet, and copy the wallet into the standard location under /u01/app/oracle/admin/tls_wallet
3. Configure GI, listeners and sqlnet.ora to use this wallet to encrypt traffic.
4. Bounce any running databases to pick up the TLS configuration.
5. Deploy the root certificates to the applications/Oracle clients that need to connect to the database.
6. Onboard new TLS connection strings to OUD for the database.

The dell_orapki_auto.sh tool has been developed to automate steps 1 and 2, but the SNOW process and downloading the signed certificate is a manual process. See section:

- Wallet Creation/Certificate Request Process

The dell_oracle_tls_auto.sh has been developed to automate step 3 (for L-DELL database servers only), but manual steps are also listed in this document should you wish to do the configuration manually. There is also a quick TLS configuration which uses a default DB engineering wallet. See the following sections:

- Quick TLS Configuration
- Automated Oracle GRID/Database Server Configuration
- Manual Oracle GRID/Database Server Configuration

Note the DB engineering wallet should only be used temporarily until an application/environment specific wallet can be created.

For step 5 there are sample client wallets available for Oracle Client versions 12.1, 12.2 and 19c, and the root certificates are available.

For L-EMC servers use dell_orapki_auto.sh for steps 1 and 2, and then follow the “Manual Oracle GRID/Database Server Configuration” section. Note that L-EMC servers might have a custom DB listener with a name LISTENER_DBNAME. This listener should be configured for port 1523/TCPS.

Quick TLS Configuration

All new Oracle server/VM builds since the 29th April 2020 will have a default DB engineering wallet installed (from /misc/software/database/oracle/tools/wallet/server/19). This contains a user certificate which expires on date "Apr 29 15:30:24 2022". No further configuration is required.

You can check if the server has this configuration/wallets by running the following status command:

```
dell_oracle_tls_auto.sh -status
```

If it does then in part of the output it will show the user and trusted certificate information:

```
PASS: User certificate is valid for at least 90 days and expires on date Apr 29 15:30:24 2022 GMT.
HELP: TNS connection strings should contain
(SECURITY=(SSL_SERVER_CERT_DN="CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies"))
PASS: Trusted certificate expires on Nov 14 21:36:35 2025 GMT with DN: CN=Dell Technologies Issuing CA
101,O=Dell Technologies,L=Round Rock,ST=Texas,C=US
PASS: Trusted certificate expires on Jul 23 17:17:44 2043 GMT with DN: CN=Dell Technologies Root
Certificate Authority 2018,OU=Cybersecurity,O=Dell Technologies,L=Round Rock,ST=Texas,C=US
PASS: Wallet contains 2 trusted certificates including a root certificate.
PASS: Wallet contains a root certificate.
```

Note the distinguished name (DN):

CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies

This is a perfectly valid certificate, and can be used to connect to any database running on the server e.g. using a TNS string like the following:

```
DBUP1D_TLS =
  (DESCRIPTION =
     (CONNECT_TIMEOUT=20)(RETRY_COUNT=3)(RETRY_DELAY=1)(TRANSPORT_CONNECT_TIMEOUT=3 SEC)
     (ADDRESS =(PROTOCOL=TCPS)(HOST=alan2de2dbscn.us.dell.com)(PORT=1523))
     (CONNECT_DATA =
       (SERVER=dedicated)
       (SERVICE_NAME=dbup1d.dev.amer.dell.com))
     (SECURITY= (SSL_SERVER_CERT_DN="CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell
Technologies")))
```

This wallet is primarily used for DDC DBaaS Oracle VM's, so that developers can access the database using TLS from day 1, but is can be used temporarily for any development environment, until a proper certificate is installed.

It should never be used for production environments as that is against Dell's security standards.

On an existing environment with no TLS enabled and with no wallet in /u01/app/oracle/admin/tls_wallet, the TLS configuration and this wallet will be installed by just running:

```
dell_oracle_tls_auto.sh -grid
```

(any running databases will need restarted manually to pick up the sqlnet.ora TLS configuration)

At this point the database servers (single node or RAC) will be fully TLS enabled. Any running databases would be configured for TLS using this (this basically sets the local_listener and remote listener parameters:

```
dell_oracle_tls_auto.sh -db <<dbshortname>> -oh $ORACLE_HOME -rolling
```

The remaining steps at this point would be:

1. Follow the 'Wallet Creation/Certificate Request Process' section to create a new wallet with a certificate specific to your application.
2. Follow the 'Updating the Server Certificate on a Running Database' section to replace the existing default wallet.

Wallet Creation/Certificate Request Process

Create a Certificate Signing Request (CSR)

The $GRID_HOME/bin/orapki command can be used to create a certificate signing request (CSR), but in order to simplify the procedure, there is a DB engineering utility in /misc/software/oracle/tools/maintenance which automates many of the steps.

These are the steps to create a CSR:

1. Create a database wallet.
2. Create a new private key which will be used to encrypt all traffic.
3. Create a certificate signing request (CSR).

The dell_orapki_auto.sh tool automates these 3 steps and can be run as follows:

```
dell_orapki_auto.sh -gencsr
```

This will generate a certificate signing request using the following defaults:

-wallet – A wallet location of /home/oracle/dell_orapki/wallet_YYYYMMDD_HH24MISS i.e. based on the date/time of the server.

-pwd – A wallet password of change1t

-cn – A CN is based on the hostname of the server. For RAC this should cluster name or default DB service name.

-certtype – rsa

Alternatively, you can specify these values via the command line and change the defaults:

```
dell_orapki_auto.sh -gencsr -wallet $HOME/dell_certs -pwd change1t -cn
ausulssd2db01.us.dell.com -certtype rsa
```

The CN is probably the most important one here, as it is the name you will give the certificate, and will also be used when a client connects to the database. Typically, this will be the database name, cluster name or server name.

For production, the wallet will be shared across production and DR clusters, so use the fully qualified database name i.e. including the .prd.amer.dell.com suffix, and choose a more complex password.

The certtype can be eca, but unfortunately this isn't supported by the Dell Certificate Authority, so always use rsa. The program will default to rsa if you don't specify it.

So start by logging onto the server, create the directory where you wish to store the wallet that will contain the certificate request e.g. $HOME/dell_certs (or leave empty to use the default), and run the following command:

```
dell_orapki_auto.sh -gencsr -cn engdloralandb06.us.dell.com -pwd change1t
```

For production please use something other than the default change1t password above.

If the command runs successfully, you'll see an output like the following:

```
[START] Running dell_orapki_auto.sh certificate request/import program.
[INFO]  Log file /tmp/dell_orapki_auto_20190520_064129.log
....
[INFO] Oracle Wallet /home/oracle/dell_orapki/wallet_20190520_064129 contains the
following:
Requested Certificates:
Subject:        CN=engdloralandb06.us.dell.com,OU=IEO,O=Dell Inc.
User Certificates:
```

```
Trusted Certificates:
Null message body; hope that's ok
[SUCCESS]  --------------------------------------------------------------
[SUCCESS]  Exit Message:  Certificate signing request successfully extracted to file
/home/oracle/dell_orapki/wallet_20190520_064129/engdloralandb06.us.dell.com.csr .
Copy the contents into the certificate signing portal.
[SUCCESS]  Log is located at /tmp/dell_orapki_auto_20190520_064129.log
[SUCCESS]  EXITING dell_orapki_auto.sh at Mon May 20 06:41:35 CDT 2019
[SUCCESS]  --------------------------------------------------------------
```

So the output in in the wallet directory. If you list the contents, you will see the following:

```
$ ls -l /home/oracle/dell_orapki/wallet_20190520_064129
total 12
-rw------- 1 oracle oinstall 2301 May 20 06:41 cwallet.sso
-rw------- 1 oracle oinstall    0 May 20 06:41 cwallet.sso.lck
-rw------- 1 oracle oinstall  967 May 20 06:41 engdloralandb06.us.dell.com.csr
-rw------- 1 oracle oinstall 2256 May 20 06:41 ewallet.p12
-rw------- 1 oracle oinstall    0 May 20 06:41 ewallet.p12.lck
```

The files are as follows:

1. engdloralandb06.us.dell.com.csr – this is the certificate signing request, and is the file that you submit to the certificate authority for signing.
2. cwallet.sso – This is a single sign on wallet i.e. the autologon wallet that doesn't need a password to access the trusted certificates and public key (the cwallet.sso.lck file is created during the creation of the wallet, but isn't actually needed).
3. ewallet.p12 – This is the password protected Oracle wallet. It is a wallet in PKCS#12 format, which is an industry standard format for storing the server, intermediate and root CA certificates, and the private key.

If you wish to query the contents of the wallet used the orapki command:

```
$ orapki wallet display -wallet .
Oracle PKI Tool : Version 12.2.0.1.0
Copyright (c) 2004, 2016, Oracle and/or its affiliates. All rights reserved.

Requested Certificates:
Subject:        CN=engdloralandb06.us.dell.com,OU=IEO,O=Dell Technologies
User Certificates:
Trusted Certificates:
```

At this stage the wallet just contains the CSR (Requested Certificate), but doesn't contain any User Certificates (Server certificate), or Trusted Certificates (e.g. the Dell CA root certificate).

If you look at the output file for the CSR, it just looks like it contains random characters:

```
$ cat engdloralandb06.us.dell.com.csr
-----BEGIN NEW CERTIFICATE REQUEST-----
MIIClTCCAX0CAQAwUDEaMBgGA1UEChMRRGVsbCBUZWNobm9sb2dpZXMxDDAKBgNV
BAsTA0lFTzEkMCIGA1UEAxMbZW5nZGxvcmFsYW5kYjA2LnVzLmRlbGwuY29tMIIB
IjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAowB0A+j5MVRKgVNI4z/AvZ3i
Jypm3QAz83AGZGGugu657m/suWwEFbZG7UEwrnduKnKd7iMnNdrdwl3oIzbh4wNn
EYozL0W1CnnohjGjzhuTvVKX5TvwIfbATzQSuJY31dFunIpYDdPyZ8rfE1QSS4mK
2Lt7gHUTg/TuDIkST0Lmj6t2wWv2l9wSseG5TW+F4YVjb4axYci+ul3V6Z81L/Zp
EyNQ7ffnou8MPPxZXQuCRA2fBYyBI+gaUBN1/8ceZO57DH42AO+62wkX5TUZuXKy
qluzR6DfwMwcUtjBoHv0mlyafkuMtna09xU0o7bUDJBh4XjE1lW6cpT2XODuFQID
```

```
AQABoAAwDQYJKoZIhvcNAQELBQADggEBABRweSPvoxHUe9lGRL4QW3skUXIeaXj/
saVhdvPEe6DTIS/1+ZxJgeC4B/dh8ULLiifmEimWh4k69qgfoFEAARHXTglcxKYR
5lennAIyUHzx2OSwdzrNsRroL1ecWwXolp6hfcBvryKlpOs7ffMcJ3bAfOOHrPZL
u0XEtM7ZHvQFy1vJE02Wjuf3r6cJZ3IxeWQZMfSoUJ4fwPgLrWllrYWjuvJu1+Z4
YtTOgQC8E8xu5b3E2D0AKFM7x/ieO0I/NM953s/5tvwagaSOiGgKO9X/T0a6kmzx
tsl4/zdcIv7Hj6/TY9zbyArD/IRFpBaKBIpyLrlk8qrcQZgJvj2NEQ0=
-----END NEW CERTIFICATE REQUEST-----
```

It will be this text that is pasted into the ServiceNow request for requesting a certificate.

You can also use openssl to view the certificate in a non-scrambled form, to check the Subject name:

```
$ openssl req -in engdloralandb06.us.dell.com.csr -text -noout
Certificate Request:
    Data:
        Version: 0 (0x0)
        Subject: O=Dell Technologies, OU=IEO, CN=engdloralandb06.us.dell.com
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
                    00:a3:00:74:03:e8:f9:31:54:4a:81:53:48:e3:3f:
                    c0:bd:9d:e2:27:2a:66:dd:00:33:f3:70:06:64:61:
                    ae:82:ee:b9:ee:6f:ec:b9:6c:04:15:b6:46:ed:41:
                    30:ae:77:6e:2a:72:9d:ee:23:27:35:da:dd:c2:5d:
                    e8:23:36:e1:e3:03:67:11:8a:33:2f:45:b5:0a:79:
                    e8:86:31:a3:ce:1b:93:bd:52:97:e5:3b:f0:21:f6:
                    c0:4f:34:12:b8:96:37:d5:d1:6e:9c:8a:58:0d:d3:
                    f2:67:ca:df:13:54:12:4b:89:8a:d8:bb:7b:80:75:
                    13:83:f4:ee:0c:89:12:4f:42:e6:8f:ab:76:c1:6b:
                    f6:97:dc:12:b1:e1:b9:4d:6f:85:e1:85:63:6f:86:
                    b1:61:c8:be:ba:5d:d5:e9:9f:35:2f:f6:69:13:23:
                    50:ed:f7:e7:a2:ef:0c:3c:fc:59:5d:0b:82:44:0d:
                    9f:05:8c:81:23:e8:1a:50:13:75:ff:c7:1e:64:ee:
                    7b:0c:7e:36:00:ef:ba:db:09:17:e5:35:19:b9:72:
                    b2:aa:5b:b3:47:a0:df:c0:cc:1c:52:d8:c1:a0:7b:
                    f4:9a:5c:9a:7e:4b:8c:b6:76:b4:f7:15:34:a3:b6:
                    d4:0c:90:61:e1:78:c4:d6:55:ba:72:94:f6:5c:e0:
                    ee:15
                Exponent: 65537 (0x10001)
        Attributes:
            a0:00
    Signature Algorithm: sha256WithRSAEncryption
         14:70:79:23:ef:a3:11:d4:7b:d9:46:44:be:10:5b:7b:24:51:
         72:1e:69:78:ff:b1:a5:61:76:f3:c4:7b:a0:d3:21:2f:f5:f9:
         9c:49:81:e0:b8:07:f7:61:f1:42:cb:8a:27:e6:12:29:96:87:
         89:3a:f6:a8:1f:a0:51:00:01:11:d7:4e:09:5c:c4:a6:11:e6:
         57:a7:9c:02:32:50:7c:f1:d8:e4:b0:77:3a:cd:b1:1a:e8:2f:
         57:9c:5b:05:e8:96:9e:a1:7d:c0:6f:af:22:a5:a4:eb:3b:7d:
         f3:1c:27:76:c0:7c:e3:87:ac:f6:4b:bb:45:c4:b4:ce:d9:1e:
         f4:05:cb:5b:c9:13:4d:96:8e:e7:f7:af:a7:09:67:72:31:79:
         64:19:31:f4:a8:50:9e:1f:c0:f8:0b:ad:69:65:ad:85:a3:ba:
         f2:6e:d7:e6:78:62:d4:ce:81:00:bc:13:cc:6e:e5:bd:c4:d8:
         3d:00:28:53:3b:c7:f8:9e:3b:42:3f:34:cf:79:de:cf:f9:b6:
```

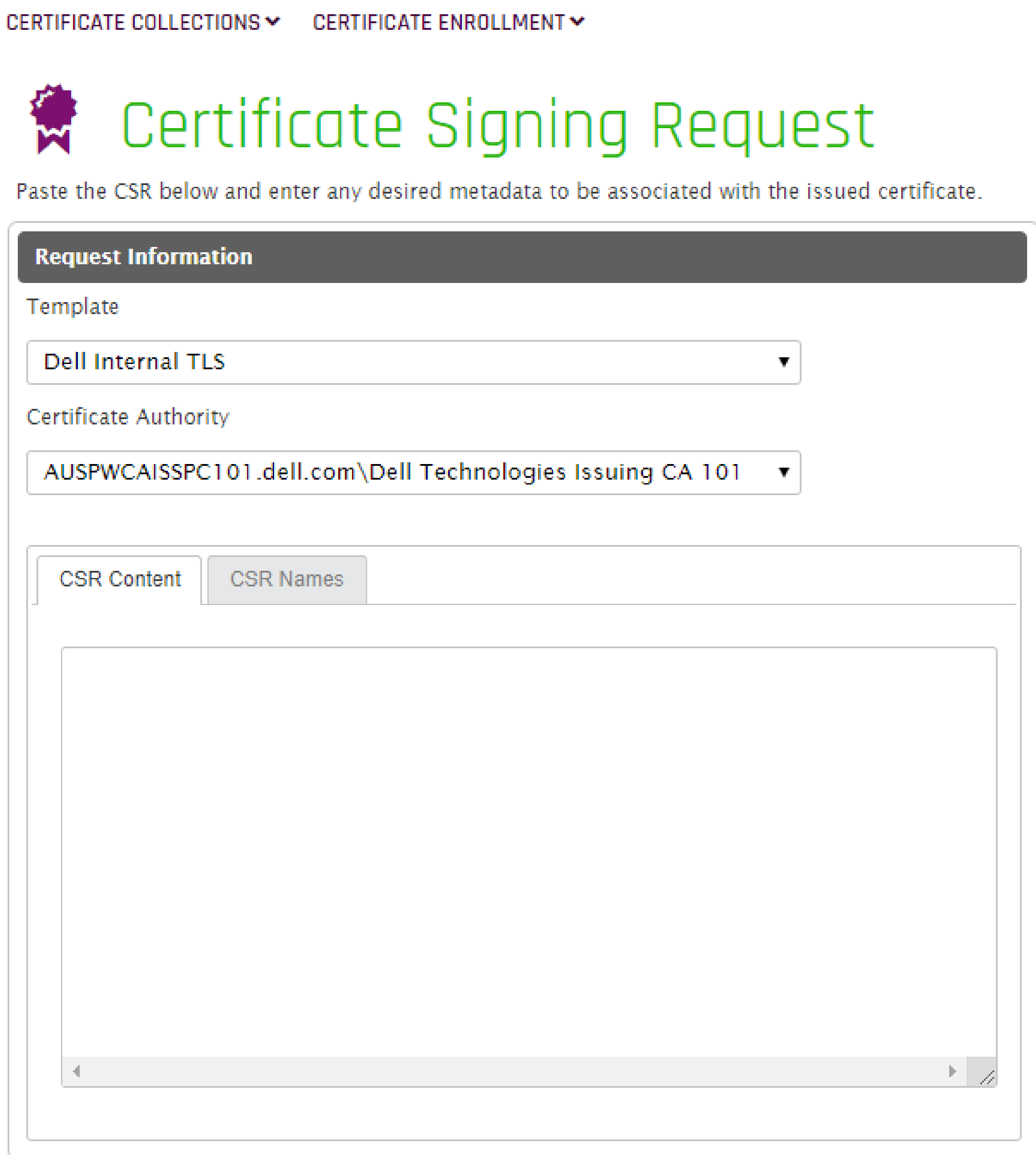
```
        fc:1a:81:a4:8e:88:68:0a:3b:d5:ff:4f:46:ba:92:6c:f1:b6:
        c9:78:ff:37:5c:22:fe:c7:8f:af:d3:63:dc:db:c8:0a:c3:fc:
        84:45:a4:16:8a:04:8a:72:2e:b9:64:f2:aa:dc:41:98:09:be:
        3d:8d:11:0d
```

Submit the Certificate Signing Request to the Dell Certificate Authority

The certificate request process is documented here. Follow that document for the latest guidance, and submit a ServiceNow request for an Internal Certificate (as databases are only ever accessed internally).

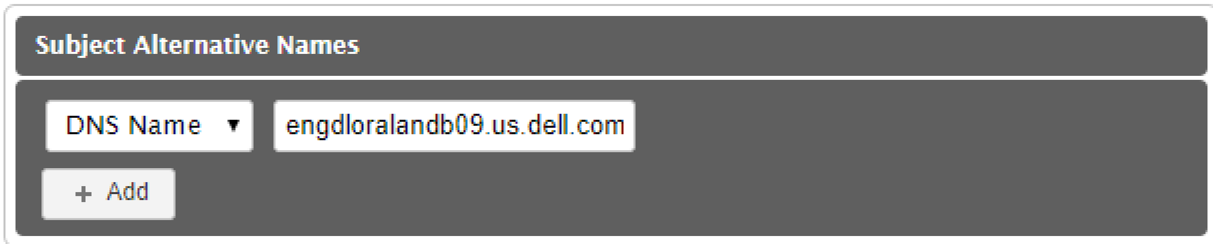
As part of that process, you will be directed to the certificate request portal here, and it is that portal where you will choose the option where you already have a CSR.

Choose the option Certificate Enrolment/CSR Enrolment and use the ‘Dell Internal TLS’ template:



Copy all the ASCII text in the base-64 encoded CSR which was generated by the dell_orapki_auto.sh tool into the ‘CSR Content‘ text box, complete the Certificate Metadata. Add the server name or scan/cname into the Subject

Alternative Names box as a DNS name (this must be the same name used in the Request ID/Common Name box in the SNOW request).



It is recommended to include a support group email address rather than an individual address, as that will ensure when the certificate is close to expire, an email warning is received by the wider group.

Leave the certificate format as PEM which is a text based format, and click Enroll.

Once submitted, complete the ServiceNow request with the request ID and submit that.

Receive Signed Certificate in PKCS#7 Format

Within 24 hours, you should receive confirmation that the request has been processed. At this point go back to the certificate portal here, and chose the Certificate Collections/Collection Manager option. Follow the instructions in the request document to collect the certificate.

Click on the certificate which matches the CN name, click on Certificate Details, and then the Download option.

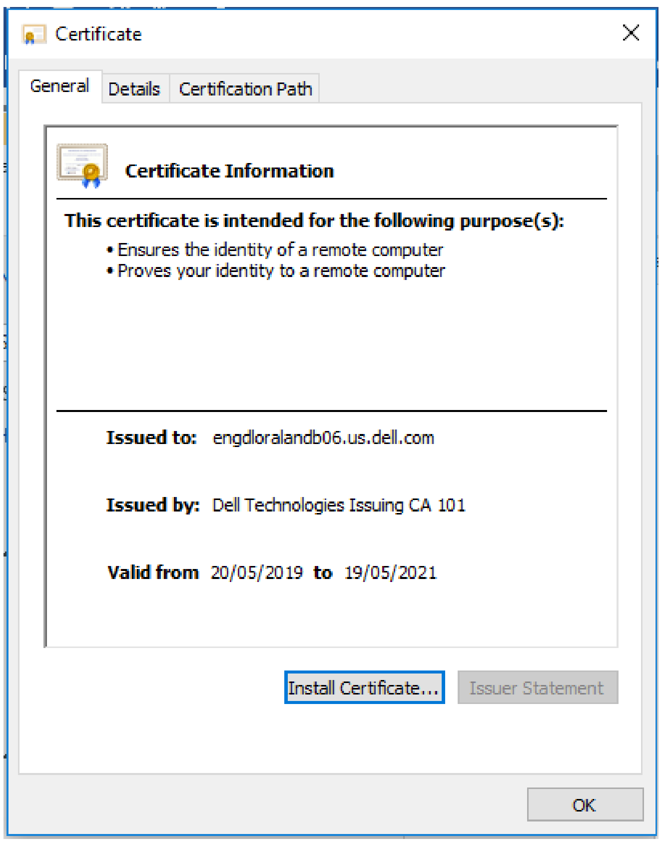
There are a few options to save the file:

- PEM encoded
- DER encoded (not available with the Include Chain option)
- P7B encoded

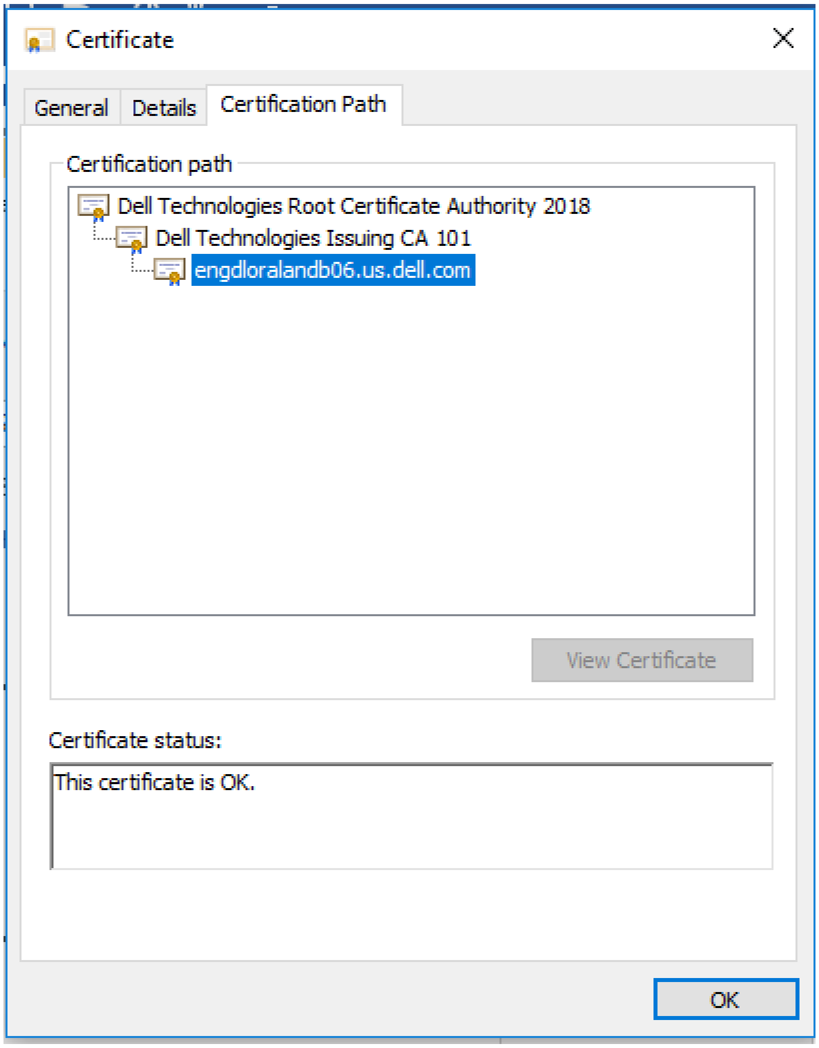
Click on the 'Include Chain' check box, and then select the PEM format and download.

It should default to a name like <servername>.pem. Additionally, use the 'Download link again, include the chain and select the P7B format. It will save the file to your PC with a name like <servername>.p7b.

Using Windows Explorer, navigate to the directory where the .pem file is located, rename the suffix from .pem to .cer and double-click on it, which should open it with the Windows Crypto Shell Extensions tool as follows:



Under the Certification Path, you can view all the certificates in the chain:



This shows 'Dell Technologies Root Certificate Authority 2018' as the root certificate in the chain.

Copy Certificate to Server

Using WinSCP, copy the <servername>.p7b file to the server, into the directory where the original CSR and Oracle wallet were.

N.B. It is possible to manually save each certificate in the chain as a separate Base-64 enclosed X.509 certificate, and then use orapki commands to import them individually into the wallet, but the dell_orapki_auto.sh script automates extracting the certificates from the p7b file into separate files, importing them in the correct order.

Import the Signed Certificate into the Oracle Wallet

In order to import a signed certificate request into a wallet using orapki, you need all the certificates in the chain. In the case of a Dell CA issued certificate, the following certificates are required:

- Dell Technologies Root Certificate Authority 2018
- Dell Technologies Issuing CA 101
- Server certificate

Note that in May 2019 the trusted certificates were no longer issued with the “Dell Root Certificate Authority 2015” CA, but changed to “Dell Technologies Root Certificate Authority 2018” CA.

orapki requires that the top level certificates be imported first, so they would need to be done in this order:

- Dell Technologies Root Certificate Authority 2018
  - Dell Technologies Issuing CA 101
    - Server certificate

This requires 3 separate files, one for each certificate.

The dell_orapki_auto.sh tool handles all this automatically, using the P7B file that was created earlier. The P7B file contains all 3 certificates, and the tool will split them into separate files (using openssl), importing them in the correct order. The tool also outputs the individual certificates as text files, as the root certificate in PEM format will be required for any Java based clients, for importing into the Java Key Store (JKS).

Log onto the server where the CSR was generated, navigate to the directory where the wallet containing the CSR is, and use WinSCP to upload the .p7b file exported in Windows to this location. Then run the dell_orapki_auto.sh command.

The following shows the command running and the output:

```
$ dell_orapki_auto.sh -importcert -wallet $HOME/dell_certs -pwd change1t -certificate certnew.p7b
[START] Running dell_orapki_auto.sh certificate request/import program.
......
[SUCCESS]  -----------------------------------------------------------
[SUCCESS]  New server wallet created in /home/oracle/dell_certs/server_wallet.
[SUCCESS]  Copy these files to the SERVER wallet directory:
[SUCCESS]      /home/oracle/dell_certs/server_wallet/cwallet.sso
[SUCCESS]      /home/oracle/dell_certs/server_wallet/ewallet.p12
[SUCCESS]
[SUCCESS]  New Oracle CLIENT wallet created in /home/oracle/dell_certs/client_wallet.
[SUCCESS]  Copy these files to the CLIENT wallet directory:
[SUCCESS]      /home/oracle/dell_certs/client_wallet/ewallet.p12
[SUCCESS]      /home/oracle/dell_certs/client_wallet/cwallet.sso
[SUCCESS]
[SUCCESS]  For Java clients, use the root certificate file for importing into client keystore:
[SUCCESS]     /home/oracle/dell_certs/DellRootCertificateAuthority2015.pem
[SUCCESS]  Example syntax for Java keytool on destination application server (default storepass is changeit):
[SUCCESS]     JRE_HOME/bin/keytool -import -alias <<certaliasname>> -file <<certificate filename>> -keystore JRE_HOME/lib/security/cacerts -storepass <<store password>>
[SUCCESS]  -----------------------------------------------------------
[SUCCESS]  -----------------------------------------------------------
[SUCCESS]  Exit Message:  Successful completion.
[SUCCESS]  Log is located at /tmp/dell_orapki_auto_20170524_082220.log
[SUCCESS]  EXITING dell_orapki_auto.sh at Wed May 24 08:22:54 CDT 2017
[SUCCESS]  -----------------------------------------------------------
```

There will be a log file created in /tmp/dell_orapki_auto_<timestamp>.log if there has been any errors.

Note the key sections of the output:

1. The SERVER wallet (the wallet which will be deployed on the database server), has been created in a sub-directory walled server_wallet. It is these wallet files that will be copied to the wallet location referenced in the $TNS_ADMIN/sqlnet.ora. The server wallet contains the following certificates:
   - Dell Technologies Root Certificate Authority 2018
   - Dell Technologies Issuing CA 101
   - Server certificate
2. The CLIENT wallet (the wallet which will be deployed to any Oracle Client connecting to the database server), has been created in a sub-directory called client_wallet. The client wallet just contains the following certificate:
   - Dell Technologies Root Certificate Authority 2018

Only the root certificate is required for any clients accessing the database server.

3. The Dell root certificate has been exported in PEM format into file DellTechnologiesRootCertificateAuthority2018.pem. This will be required for importing into the Java trust keystore, for any Java based clients such as Weblogic. This is the same certificate which is in the CLIENT wallet.

Automated Oracle GRID/Database Server Configuration

Summary

A tool has been developed which completely automates the configuration of a server or RAC cluster for TCPS traffic. It covers the following steps:

1. Configures the active GRID_HOME (Restart or RAC) to support TCPS (TLS) connections.
2. Optionally creates a wallet containing a self signed certificate in directory /u01/app/oracle/admin/tls_wallet if no wallet already exists in that directory. This isn't suitable for general use, but it allows validation of the TLS configuration.
3. Optionally configures the database home sqlnet.ora file using a symbolic link to point to the GRID_HOME TNS_ADMIN sqlnet.ora file.
4. Optionally configures the database local_listener parameter.
5. Optionally rolling restarts the listeners.
6. Generate a summary of the TLS configuration to confirm everything is configured correctly.

Due to a bug on Oracle Restart, the listener.ora file might need to be manually corrected, if the GRID_HOME is missing a patch for bug 22144696. This patch is included in the April 2019 PSU for 12.1 and 12.2. The April 2019 PSU can be found in the following location, but note it isn't part of the yearly PSU certification, so if you choose to apply it ensure it is tested.

```
/misc/software/oracle/12cR2/12.2.0.1/CPUPatches/ruapr2019
```

Rather than apply an untested PSU, it is often just better to add the additional line manually to the LISTENER configuration in the $TNS_ADMIN/listener.ora file for the node e.g.

```
(ADDRESS = (PROTOCOL = TCPS)(HOST = <<hostname or IP>>)(PORT = 1523))
```

The rest of this section includes the instructions for running the automated configuration tool.

Copy the Certificates to /u01/app/oracle/admin/tls_wallet

If the previous step to generate a CSR and request a certificate from the Dell Certificate Authority has been completed, there will be 2 wallet files under a directory like /home/oracle/dell_certs/server_wallet (or another directory depending on the location the wallet was generated in).

Copy these files to /u01/app/oracle/admin/tls_wallet e.g. create it and copy the files:

```
mkdir /u01/app/oracle/admin/tls_wallet
cp /home/oracle/dell_certs/server_wallet/* /u01/app/oracle/admin/tls_wallet
```

Configure GRID_HOME for TCPS.

Run the following command to configure the GRID_HOME (Restart or RAC) for TCPS on the first node of the cluster:

```
dell_oracle_tls_auto.sh -grid
```

If all goes well the output should be like the following (this assumes no wallet exists under directory / u01/app/oracle/admin/tls_wallet, but if the directory contains valid wallet files it will skip the generation of a self signed wallet and just use the generated wallet):

```
[START] Running dell_oracle_tls_auto.sh TLS Oracle configuration program.
[INFO]  Log file /tmp/dell_oracle_tls_auto_20190618_053431.log
[START] Running dell_orapki_auto.sh certificate request/import program.
[INFO]  Log file /tmp/dell_orapki_auto_20190618_053432.log
.......
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  New server wallet created in
/u01/app/oracle/admin/tls_wallet/server_wallet.
[SUCCESS]  Copy these files to the SERVER wallet directory (usually
/u01/app/oracle/admin/tls_wallet):
[SUCCESS]      /u01/app/oracle/admin/tls_wallet/server_wallet/cwallet.sso
[SUCCESS]      /u01/app/oracle/admin/tls_wallet/server_wallet/ewallet.p12
[SUCCESS]
[SUCCESS]  New Oracle CLIENT wallet created in
/u01/app/oracle/admin/tls_wallet/client_wallet. The client wallet password is
welcome1
[SUCCESS]  Copy these files to the CLIENT wallet directory:
[SUCCESS]      /u01/app/oracle/admin/tls_wallet/client_wallet/ewallet.p12
[SUCCESS]      /u01/app/oracle/admin/tls_wallet/client_wallet/cwallet.sso
[SUCCESS]
[SUCCESS]  For Java clients, use the root certificate file for importing into client
keystore:
[SUCCESS]     /u01/app/oracle/admin/tls_wallet/root_jntstracs301.us.dell.com.pem
[SUCCESS]  Example syntax for Java keytool on destination application server (default
storepass is changeit):
[SUCCESS]     JRE_HOME/bin/keytool -import -alias <<certaliasname>> -file
<<certificate filename>> -keystore JRE_HOME/lib/security/cacerts -storepass <<store
password>>
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  Exit Message:  Successful completion.
[SUCCESS]  Log is located at /tmp/dell_orapki_auto_20190618_053432.log
[SUCCESS]  EXITING dell_orapki_auto.sh at Tue Jun 18 05:34:56 CDT 2019
[SUCCESS]  ---------------------------------------------------------------
```

```
cp: cannot stat '/u01/app/oracle/admin/tls_wallet/cwallet.sso': No such file or
directory
cp: cannot stat '/u01/app/oracle/admin/tls_wallet/ewallet.p12': No such file or
directory
cwallet.sso
100% 4109     1.3MB/s   00:00
ewallet.p12
100% 4064     1.1MB/s   00:00
[START] Running dell_oracle_tls_auto.sh TLS Oracle configuration program.
[INFO]  Log file /tmp/dell_oracle_tls_auto_20190618_053505.log

[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  Exit Message:  Successful completion on node jntstracs302.
[SUCCESS]  Log is located at /tmp/dell_oracle_tls_auto_20190618_053505.log
[SUCCESS]  EXITING dell_oracle_tls_auto.sh at Tue Jun 18 05:35:07 CDT 2019
[SUCCESS]  ---------------------------------------------------------------

[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  ---------------------------------------------------------------
[SUCCESS]  Exit Message:  Successful completion on node jntstracs301.
[SUCCESS]  Log is located at /tmp/dell_oracle_tls_auto_20190618_053431.log
[SUCCESS]  EXITING dell_oracle_tls_auto.sh at Tue Jun 18 05:35:24 CDT 2019
[SUCCESS]  ---------------------------------------------------------------
```

Run this afterwards to confirm everything was configured correctly:

```
dell_oracle_tls_auto.sh -status
```

If all is okay then you are good to go.

If the wallet directory did not contain a valid wallet, then the program will create a wallet with a self signed certificate. While that is acceptable for a simple test, it is important to generate a certificate using a trusted certificate authority, in this case Dell's certificate authority, and copy the wallets into the tls_wallet directory, as self signed certificates are not considered secure.

Configure ORACLE_HOME and database for TCPS.

For an Oracle Restart database, run the following command to configure the database home and database <<dbshortname>>:

```
dell_oracle_tls_auto.sh -db <<dbshortname>> -oh $ORACLE_HOME -rolling
```

For an Oracle RAC database, run the following command on the first node of the cluster for database <<dbshortname>>:

```
dell_oracle_tls_auto.sh -db <<dbshortname>> -oh $ORACLE_HOME -rolling
```

This should automatically pick up the DB_UNIQUE_NAME by running an SRVCTL command and will then configure the database parameters to support TCPS.

As the local listener parameter is configured on each node, it restarts the local listener. For an Oracle RAC cluster, once all nodes are configured it does a rolling restart of the scan listeners too (remove the -rolling option to avoid the listener restarts if required).

Validate the configuration

Run the following command to validate the configuration:

```
dell_oracle_tls_auto.sh -status
```

It should output something like the following

```
$ dell_oracle_tls_auto.sh -status
[START] Running dell_oracle_tls_auto.sh TLS Oracle configuration program.
[INFO]  Log file /tmp/dell_oracle_tls_auto_20190711_160538.log
        ==================================================================
        TLS Configuration status
        ==================================================================

         INFO: **** Node engdloralandb09: Check sqlnet.ora ***
         PASS: Node engdloralandb09: File
/u01/app/grid/product/19.0.0.0/grid_1/network/admin/sqlnet.ora exists.
         PASS: Node engdloralandb09: Parameter WALLET_LOCATION is set in file
/u01/app/grid/product/19.0.0.0/grid_1/network/admin/sqlnet.ora
         PASS: Node engdloralandb09: Parameter SSL_CLIENT_AUTHENTICATION is set to
FALSE in file /u01/app/grid/product/19.0.0.0/grid_1/network/admin/sqlnet.ora
         PASS: Node engdloralandb09: Parameter SSL_SERVER_DN_MATCH is set to ON in
file /u01/app/grid/product/19.0.0.0/grid_1/network/admin/sqlnet.ora
         PASS: Node engdloralandb09: Parameter SSL_VERSION is set to 1.2 in file
/u01/app/grid/product/19.0.0.0/grid_1/network/admin/sqlnet.ora
         PASS: Node engdloralandb09: WALLET_LOCATION is set in sqlnet.ora to
/u01/app/oracle/admin/tlss_wallet
         PASS: Symbolic link
/u01/app/oracle/product/19.0.0.0/db_1/network/admin/sqlnet.ora exists

         INFO: **** Node engdloralandb09: Check listener.ora ***
         PASS: File /u01/app/grid/product/19.0.0.0/grid_1/network/admin/listener.ora
exists.
         PASS: Node engdloralandb09: Parameter WALLET_LOCATION is set in file
/u01/app/grid/product/19.0.0.0/grid_1/network/admin/listener.ora
         PASS: Node engdloralandb09: Parameter SSL_CLIENT_AUTHENTICATION is set to
FALSE in file /u01/app/grid/product/19.0.0.0/grid_1/network/admin/listener.ora
         PASS: Node engdloralandb09: Parameter SSL_VERSION is set to 1.2 in file
/u01/app/grid/product/19.0.0.0/grid_1/network/admin/listener.ora

         INFO: **** Check GI TCPS/1523 listener configuration ***
         PASS: listener is configured for TCPS/port 1523

         INFO: **** Check listeners to see if they are listening on for tcps/port
1523 ***
         PASS: Node engdloralandb09: LISTENER is listening for TCPS requests on port
1523.

         INFO: **** Check running databases for local_listener configuration ****
         PASS: Node engdloralandb09 Database: dbup1d - local_listener parameter
contains an ADDRESS entry for TCPS and 1523.

         INFO: **** Check wallet ***
         PASS: Node engdloralandb09: Wallet directory
/u01/app/oracle/admin/tlss_wallet exists.
```

```
         PASS: Node engdloralandb09: Wallet files cwallet.sso and ewallet.p12 exist.
         PASS: User certificate expires on Jul  1 17:36:19 2029 GMT with DN:
CN=engdloralandb09.us.dell.com,OU=IEO,O=Dell Technologies
                 TNS connection strings should contain
(SECURITY=(SSL_SERVER_CERT_DN="CN=engdloralandb09.us.dell.com,OU=IEO,O=Dell
Technologies"))
         PASS: Trusted certificate expires on Jul  1 17:36:19 2029 GMT with DN:
CN=engdloralandb09.us.dell.com,OU=IEO,O=Dell Technologies
         FAIL: Wallet should contain at least 2 trusted certificates, including a
root and intermediate certificate.
         FAIL: Wallet does not contain a trusted root certificate.
         FAIL: Wallet contains a self-signed certificate. Certificates must be issued
by the Dell Certificate Authority.
[SUCCESS]  ------------------------------------------------------------
[SUCCESS]  Exit Message:  Successful completion on node engdloralandb09.
[SUCCESS]  Log is located at /tmp/dell_oracle_tls_auto_20190711_160538.log
[SUCCESS]  EXITING dell_oracle_tls_auto.sh at Thu Jul 11 16:05:51 CDT 2019
[SUCCESS]  ------------------------------------------------------------
```

In this example it highlights that the certificates are self-signed and not trusted, but if the CSR process was followed these checks should pass.

On a RAC cluster, if there's any indication that the scan listeners have blocked connections, then the database will need bounced (even on a single instance Oracle Restart environment the database is likely to need restarted). Begin by restarting the listeners on each node, and if that doesn't resolve the blocked connection issue then stop and start the database on all nodes.

The output should show something like the following if all is okay:

```
         INFO: **** Check listeners to see if they are listening on for tcps/port
1523 ***
         PASS: Node engdlorala7d01: LISTENER is listening for TCPS requests on port
1523.
         PASS: Node engdlorala7d01: LISTENER_SCAN2 is listening for TCPS requests on
port 1523.
         PASS: Node engdlorala7d01: LISTENER_SCAN2 has no blocked states for any
database service.
         PASS: Node engdlorala7d01: LISTENER_SCAN3 is listening for TCPS requests on
port 1523.
         PASS: Node engdlorala7d01: LISTENER_SCAN3 has no blocked states for any
database service.
         PASS: Node engdlorala7d02: LISTENER is listening for TCPS requests on port
1523.
         PASS: Node engdlorala7d02: LISTENER_SCAN1 is listening for TCPS requests on
port 1523.
         PASS: Node engdlorala7d02: LISTENER_SCAN1 has no blocked states for any
database service.
```

Manual Oracle GRID/Database Server Configuration

Skip this section if you followed the automated configuration section, or use it to help troubleshoot any configuration issues.

Create Wallet Directory and Copy Server Wallet

The server wallet should be placed under a directory under the /u01/app/oracle/admin/tls_wallet. Previous versions of this guide placed the wallet under $TNS_ADMIN/wallet, but as this directory is under GRID_HOME, and GRID_HOME will be rolling patched with new versions, the wallet has been moved outside that location. If there is a need to have separate wallets for a server with multiple databases (perhaps if each database has its own listener), then create the wallet folder under /u01/app/oracle/admin/$ORACLE_DB/tls_wallet, but the recommendation is that the wallet is common to all databases on the server, as the TLS configuration is done at the listener level, which runs out of the GRID_HOME and not the database home.

Create the directory as follows:

```
mkdir -p /u01/app/oracle/admin/tls_wallet
```

Now copy the server wallet to this directory, and restrict the permissions to 750 to ensure that only oracle can view the wallet:

```
cp $HOME/dell_certs/server_wallet/cwallet.sso /u01/app/oracle/admin/tls_wallet
cp $HOME/dell_certs/server_wallet/ewallet.p12 /u01/app/oracle/admin/tls_wallet
chmod -R 750 /u01/app/oracle/admin/tls_wallet
```

Repeat this process on each node of the cluster.

N.B. Please note that this wallet directory MUST BE different to the wallet directory used by TDE. TLS uses the sqlnet.ora parameter WALLET_LOCATION to identify the location of the TLS wallet. TDE uses the sqlnet.ora parameter ENCRYPTION_WALLET_LOCATION to identify the location of the TDE wallet. If ENCRYPTION _WALLET_LOCATION is not specified, TDE will fall back to using WALLET_LOCATION, and will usually fail if there are TLS certificates in the same wallet as the TDE certificate.

Typically the TDE wallet will be under $ORACLE_BASE/admin/$ORACLE_DB/*wallet*, which is why we have put the TLS wallet under a different directory $ORACLE_BASE/admin/***tls****_wallet.*

Further information can be found in these Metalink Notes:

- The Impact of the Sqlnet Settings on Database Security (sqlnet.ora Security Parameters and Wallet Location) (Doc ID 1240824.1)
- Unable To Open TDE Encryption Wallet After Applying MES Bundle Patch or After importing SSL certificates into TDE wallet (Doc ID 2192475.1)

Local Listener Configuration

There are 1 or more listeners running on an Oracle database server. On single node environments there will be just a single standard listener called 'LISTENER'. On RAC environments there will be 3 additional listeners for each SCAN IP. This section describes the configuration of the standard LISTENER and applies to both standalone and RAC database servers. On L-EMC servers there will be a custom LISTENER_DBNAME so that is the listener which should be configured for TCPS.

Configuring TLS for a GRID listener on a Dell standard database VM, is slightly different to the configuration on a physical RAC cluster, as Oracle Restart on a standalone VM database server doesn't seem to handle the TCPS listener.

Start by adding the wallet location to the $TNS_ADMIN/sqlnet.ora file, by appending the following lines to the existing file:

```
# Disable SSL client authentication
SSL_CLIENT_AUTHENTICATION = FALSE

# Check server DN to ensure it is correct
SSL_SERVER_DN_MATCH=ON

# Force TLS v1.2
SSL_VERSION=1.2

# Set the wallet location
WALLET_LOCATION =
  (SOURCE =
    (METHOD = FILE)
    (METHOD_DATA =
      (DIRECTORY = /u01/app/oracle/admin/tls_wallet)
    )
  )

# Set the approved Dell cipher suites
SSL_CIPHER_SUITES=(SSL_ECDHE_RSA_WITH_AES_128_GCM_SHA256,SSL_ECDHE_RSA_WITH_AES_256_G
CM_SHA384,SSL_ECDHE_RSA_WITH_AES_256_CBC_SHA384,SSL_ECDHE_RSA_WITH_AES_128_CBC_SHA256
,SSL_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,SSL_ECDHE_ECDSA_WITH_AES_128_CBC_SHA,SSL_ECD
HE_ECDSA_WITH_AES_128_CBC_SHA256,SSL_ECDHE_ECDSA_WITH_AES_256_CBC_SHA,SSL_ECDHE_ECDSA
_WITH_AES_256_CBC_SHA384)
```

Amend the DIRECTORY entry if the wallet was placed elsewhere.

Now append the same SSL_CLIENT_AUTHENTICATION, SSL_VERSION, WALLET_LOCATION and SSL_CIPHER_SUITES information to the existing $TNS_ADMIN/listener.ora file (SSL_SERVER_DN_MATCH isn't required as this is a client setting).

Note: There have been cases when putting the WALLET_LOCATION on multiple lines in the listener.ora hasn't been recognised, so if there are issues then put WALLET_LOCATION on a single line.

Note for 12.1 Client: The SSL_CIPHER_SUITES listed here are required by the Dell cipher security standard which is documented here. Oracle Client 12.1 doesn't support these ciphers, unless it is patched with October 2018 PSU or later, or patch 18604692 is applied. If the application uses the 12.1 client, then the team should really upgrade to the 19c client. If that isn't possible then ask them to patch the client with the latest PSU. If that isn't possible, then comment out the cipher parameter in the listener.ora and sqlnet.ora. See metalink note 2194424.1 for list of supported cipher suites for each version of Oracle.

Now let's capture the status of the listener by running lsnrctl status e.g.

```
$ lsnrctl status
LSNRCTL for Linux: Version 12.1.0.2.0 - Production on 25-MAY-2017 09:47:30

Copyright (c) 1991, 2014, Oracle.  All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER)))
STATUS of the LISTENER
------------------------
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 12.1.0.2.0 - Production
Start Date                11-MAY-2017 09:08:08
Uptime                    14 days 0 hr. 39 min. 21 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File
/u01/app/grid/product/12.1.0.2/grid_1/network/admin/listener.ora
Listener Log File
/u01/app/oracle/diag/tnslsnr/ausdlssd2db01/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=LISTENER)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=10.161.8.27)(PORT=1521)))
Services Summary...
Service "+ASM" has 1 instance(s).
  Instance "+ASM", status READY, has 1 handler(s) for this service...
Service "ssd2d.dev.amer.dell.com" has 1 instance(s).
  Instance "ssd2d", status READY, has 1 handler(s) for this service...
The command completed successfully
```

You can see that the listener is only running on port 1521 at this point.

Now let's run the srvctl configuration command to check what GRID has stored:

```
$ srvctl config listener
Name: LISTENER
Type: Database Listener
Home: /u01/app/grid/product/12.1.0.2/grid_1
End points: TCP:1521
Listener is enabled.
```

That also confirms that the GRID configuration just has port 1521.

Now let's add the 1523 endpoint to the GRID listener configuration, and then run the config command again:

```
srvctl modify listener -p "TCP:1521/TCPS:1523"
$ srvctl config listener
Name: LISTENER
Type: Database Listener
Home: /u01/app/grid/product/12.1.0.2/grid_1
End points: TCP:1521/TCPS:1523
Listener is enabled.
```

It now shows port 1523 is there too, using TCPS protocol.

Stop and start the listener:

```
srvctl stop listener; srvctl start listener
```

Run the lsnrctl status command again, to check that the listener is now showing both ports 1521 and 1523:

```
$ lsnrctl status
LSNRCTL for Linux: Version 12.1.0.2.0 - Production on 25-MAY-2017 09:47:30

Copyright (c) 1991, 2014, Oracle.  All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER)))
STATUS of the LISTENER
------------------------
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 12.1.0.2.0 - Production
Start Date                11-MAY-2017 09:08:08
Uptime                    14 days 0 hr. 39 min. 21 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File
/u01/app/grid/product/12.1.0.2/grid_1/network/admin/listener.ora
Listener Log File
/u01/app/oracle/diag/tnslsnr/ausdlssd2db01/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=LISTENER)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=10.161.8.27)(PORT=1523)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=10.161.8.27)(PORT=1521)))
Services Summary...
Service "+ASM" has 1 instance(s).
  Instance "+ASM", status READY, has 1 handler(s) for this service...
Service "ssd2d.dev.amer.dell.com" has 1 instance(s).
  Instance "ssd2d", status READY, has 1 handler(s) for this service...
The command completed successfully
```

On Oracle Restart VM's, there is a bug (22144696) which stops GRID registering this 2nd port 1523 listener, in which case you will need to edit the $GRID_HOME/network/admin/listener.ora file to manually add the entry. The bug is fixed in the April 2019 PSU's for 12.1 and 12.2.

Here is an example of an updated file:

```
LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = IPC)(KEY = LISTENER))
      (ADDRESS = (PROTOCOL = TCP)(HOST = ausdlssd2db01.us.dell.com)(PORT = 1521))
      (ADDRESS = (PROTOCOL = TCPS)(HOST = ausdlssd2db01.us.dell.com)(PORT = 1523))
    )
  )
```

Make sure you add the entry after the IPC entry. T

Now after stopping and starting the listener, you should see port 1523 as an endpoint.

Manual editing of the listener.ora file should not be needed on a RAC cluster.

Repeat the process on each node in the cluster.

SCAN Listener Configuration

On a RAC cluster there will be 3 additional SCAN listeners which need to be configured. The first stage is to check which nodes the SCAN listeners are running on:

```
$ srvctl status scan_listener
SCAN Listener LISTENER_SCAN1 is enabled
SCAN listener LISTENER_SCAN1 is running on node austldbengdb08
SCAN Listener LISTENER_SCAN2 is enabled
SCAN listener LISTENER_SCAN2 is running on node austldbengdb07
SCAN Listener LISTENER_SCAN3 is enabled
SCAN listener LISTENER_SCAN3 is running on node austldbengdb07
```

On any node where there is a scan listener running, query the status:

```
$ lsnrctl status LISTENER_SCAN3
LSNRCTL for Linux: Version 12.1.0.2.0 - Production on 25-MAY-2017 10:54:21

Copyright (c) 1991, 2014, Oracle.  All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN3)))
STATUS of the LISTENER
------------------------
Alias                     LISTENER_SCAN3
Version                   TNSLSNR for Linux: Version 12.1.0.2.0 - Production
Start Date                26-APR-2017 07:13:19
Uptime                    29 days 3 hr. 41 min. 1 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File
/u01/app/grid/product/12.1.0.2/grid_1/network/admin/listener.ora
Listener Log File
/u01/app/oracle/diag/tnslsnr/austldbengdb07/listener_scan3/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=LISTENER_SCAN3)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=10.160.120.228)(PORT=1521)))
Services Summary...
Service "cln" has 1 instance(s).
  Instance "cln4", status READY, has 2 handler(s) for this service...
Service "dgtst_env3.dev.amer.dell.com" has 1 instance(s).
  Instance "dgtst1", status READY, has 1 handler(s) for this service...
Service "patdb.dev.amer.dell.com" has 1 instance(s).
  Instance "patdb1", status READY, has 2 handler(s) for this service...
The command completed successfully
```

Run the following command to add the TCPS protocol on port 1523 for the scan listeners:

```
srvctl modify scan_listener -p TCP:1521/TCPS:1523
```

You can check the configuration to confirm the change as follows:

```
$ srvctl config scan_listener
SCAN Listener LISTENER_SCAN1 exists. Port: TCP:1521/TCPS:1523
Registration invited nodes:
Registration invited subnets:
SCAN Listener is enabled.
```

```
SCAN Listener is individually enabled on nodes:
SCAN Listener is individually disabled on nodes:
SCAN Listener LISTENER_SCAN2 exists. Port: TCP:1521/TCPS:1523
Registration invited nodes:
Registration invited subnets:
SCAN Listener is enabled.
SCAN Listener is individually enabled on nodes:
SCAN Listener is individually disabled on nodes:
SCAN Listener LISTENER_SCAN3 exists. Port: TCP:1521/TCPS:1523
Registration invited nodes:
Registration invited subnets:
SCAN Listener is enabled.
SCAN Listener is individually enabled on nodes:
SCAN Listener is individually disabled on nodes:
```

Now run the following to stop and start the scan listeners:

```
srvctl stop scan_listener
srvctl stop scan
srvctl start scan
srvctl start scan_listener
```

This should stop and start the scan listeners on all nodes of the cluster. If there's a need to do this in a rolling fashion to avoid a short period when there's no scan running, append the -scannumber flag to the command e.g.

```
srvctl stop scan_listener -scannumber 1
srvctl stop scan -scannumber 1
srvctl start scan -scannumber 1
srvctl start scan_listener -scannumber 1
```

And repeat for scan's 2 and 3.

Now check the status of each scan listener as follows (this needs to be run on the node where the SCAN listener is running):

```
$ lsnrctl status LISTENER_SCAN1

LSNRCTL for Linux: Version 12.1.0.2.0 - Production on 26-APR-2017 05:59:26

Copyright (c) 1991, 2014, Oracle.  All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN1)))
STATUS of the LISTENER
------------------------
Alias                     LISTENER_SCAN1
Version                   TNSLSNR for Linux: Version 12.1.0.2.0 - Production
Start Date                26-APR-2017 05:57:06
Uptime                    0 days 0 hr. 2 min. 19 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File
/u01/app/grid/product/12.1.0.2/grid_1/network/admin/listener.ora
Listener Log File
/u01/app/oracle/diag/tnslsnr/austldbengdb07/listener_scan1/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=ipc)(KEY=LISTENER_SCAN1)))
```

```
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcps)(HOST=10.160.120.229)(PORT=1523)))
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=10.160.120.229)(PORT=1521)))
Services Summary...
Service "cln" has 1 instance(s).
  Instance "cln4", status READY, has 2 handler(s) for this service...
Service "dgtst_env3.dev.amer.dell.com" has 2 instance(s).
  Instance "dgtst1", status READY, has 1 handler(s) for this service...
  Instance "dgtst2", status READY, has 1 handler(s) for this service...
Service "patdb.dev.amer.dell.com" has 2 instance(s).
  Instance "patdb1", status READY, has 2 handler(s) for this service...
  Instance "patdb2", status READY, has 2 handler(s) for this service...
The command completed successfully
```

You can see there's now a process listening for TCPS traffic on port 1523.

Database Configuration

There are 3 stages to this:

1. Ensure the database references a sqlnet.ora file that has the WALLET_LOCATION, SSL_VERSION and SSL_SERVER_DN_MATCH configured.
2. Amend the local_listener database parameter to register with both the TCP and TCPS listener ports.
3. Optionally bounce the database. This is sometimes needed if the sqlnet.ora file location wasn't correct when the database was last restarted.

The first stage is to ensure the database references the WALLET_LOCATION in the sqlnet.ora. There are 3 options:

1. Ensure that the location of the TNS_ADMIN is specified in the database configuration at the cluster level, and is pointing to $GRID_HOME/network/admin.
2. Create a sqlnet.ora file under $ORACLE_HOME/network/admin.
3. Create a symbolic link to TNS_ADMIN/sqlnet.ora under $ORACLE_HOME/network/admin.

The first option is only available on RAC clusters or Oracle Restart (which should apply to 100% of servers in Dell), so just run the following command to specify the location of TNS_ADMIN for the database:

```
srvctl setenv database -d <DBNAME> -env
"TNS_ADMIN=/u01/app/grid/product/12.1.0.2/grid_1/network/admin"
```

Amend the location to reference the correct $GRID_HOME.

If a database isn't started using srvctl, then it would be worth creating/amending $ORACLE_HOME/network/admin/sqlnet.ora file, and adding the wallet information to it:

```
# Check server DN to ensure it is correct
SSL_SERVER_DN_MATCH=ON

# Force TLS v1.2
SSL_VERSION=1.2

# Set the wallet location
WALLET_LOCATION =
  (SOURCE =
    (METHOD = FILE)
    (METHOD_DATA =
      (DIRECTORY = /u01/app/oracle/admin/tls_wallet)
    )
  )

# Set the approved Dell cipher suites
SSL_CIPHER_SUITES=(SSL_ECDHE_RSA_WITH_AES_128_GCM_SHA256,SSL_ECDHE_RSA_WITH_AES_256_G
CM_SHA384,SSL_ECDHE_RSA_WITH_AES_256_CBC_SHA384,SSL_ECDHE_RSA_WITH_AES_128_CBC_SHA256
,SSL_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,SSL_ECDHE_ECDSA_WITH_AES_128_CBC_SHA,SSL_ECD
HE_ECDSA_WITH_AES_128_CBC_SHA256,SSL_ECDHE_ECDSA_WITH_AES_256_CBC_SHA,SSL_ECDHE_ECDSA
_WITH_AES_256_CBC_SHA384)
```

This assumes that you are using the same wallet as the listener.

An alternative to creating sqlnet.ora file under the database $ORACLE_HOME/network/admin folder, is just to create a symbolic link to the TNS_ADMIN sqlnet.ora:

```
cd $ORACLE_HOME/network/admin
ln -s $TNS_ADMIN/sqlnet.ora sqlnet.ora
```

Now amend the local_listener database parameter to register its services with both port 1521 and 1523. Ensure it matches the following:

```
patdb1 SQL> show parameter local_listener

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
local_listener                       string      (DESCRIPTION=(ADDRESS_LIST=(AD
                                                 DRESS=(PROTOCOL=TCPS)(HOST=10.
                                                 160.121.27)(PORT=1523))(ADDRES
                                                 S=(PROTOCOL=TCP)(HOST=10.160.1
                                                 21.27)(PORT=1521))))
```

Note that on many environments the local_listener just uses ADDRESS entries, but this often doesn't work with TCPS, so it's recommended to use an entry containing DESCRIPTION, ADDRESS_LIST and ADDRESS e.g.

```
(DESCRIPTION=(
 (ADDRESS_LIST=(
  (ADDRESS=
   (PROTOCOL=TCP)(HOST=<host or VIP IP>)(PORT=1521)
  )
  (ADDRESS=
   (PROTOCOL=TCPS)(HOST=<host or VIP IP>)(PORT=1523)
  )
 )
)
```

An example of amending the local_listener parameter:

```
SQL> alter system set
local_listener="(DESCRIPTION=(ADDRESS_LIST=(ADDRESS=(PROTOCOL=TCPS)(HOST=10.160.121.2
7)(PORT=1523))(ADDRESS=(PROTOCOL=TCP)(HOST=10.160.121.27)(PORT=1521))))" scope=both
sid='patdb1';

System altered.

SQL> show parameter local_listener

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
local_listener                       string      (DESCRIPTION=(ADDRESS_LIST=(AD
                                                 DRESS=(PROTOCOL=TCPS)(HOST=10.
                                                 160.121.27)(PORT=1523))(ADDRES
                                                 S=(PROTOCOL=TCP)(HOST=10.160.1
                                                 21.27)(PORT=1521))))
```

On standalone database VM's, the local listener parameter might refer to a TNS alias e.g.

```
SQL> show parameter local_listener

NAME                                 TYPE        VALUE
------------------------------------ ----------- ------------------------------
local_listener                       string      LISTENER_DBNAME
```

So amend the $ORACLE_HOME/network/admin/tnsnames.ora entry for this LISTENER_<DBNAME> to:

```
LISTENER_DBNAME =
  (DESCRIPTION =
     (ADDRESS_LIST=
        (ADDRESS = (PROTOCOL = TCP) (HOST = hostname1.us.dell.com)(PORT = 1521))
        (ADDRESS = (PROTOCOL = TCPS)(HOST = hostname1.us.dell.com)(PORT = 1523))
     )
  )
```

This has to be repeated on each database node in the cluster.

This change should take effect immediately, or the next time the database updates its service information with the listener. If there are issues you might receive the following error when trying to connect to the service running TCPS on port 1523:

```
ORA-12520: TNS:listener could not find available handler for requested type of server
```

If you see this error, you might need to bounce the listeners and see if that helps. If it doesn't then you will need to restart the database.

You can check if the scan listeners are handling both TCPS and TCP connections by using the lsnrctl services command. You should look for a DEDICATED handler in state:ready, for the TCPS/1523 connection e.g.:

```
$ lsnrctl services LISTENER_SCAN2

LSNRCTL for Linux: Version 12.1.0.2.0 - Production on 08-JUN-2017 09:52:10

Copyright (c) 1991, 2014, Oracle.  All rights reserved.

Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=IPC)(KEY=LISTENER_SCAN2)))
Services Summary...
Service "patdb.dev.amer.dell.com" has 2 instance(s).
  Instance "patdb1", status READY, has 2 handler(s) for this service...
    Handler(s):
      "DEDICATED" established:0 refused:0 state:ready
         REMOTE SERVER
         (ADDRESS=(PROTOCOL=TCPS)(HOST=10.160.121.27)(PORT=1523))
      "DEDICATED" established:0 refused:0 state:ready
         REMOTE SERVER
         (ADDRESS=(PROTOCOL=TCP)(HOST=10.160.121.27)(PORT=1521))
  Instance "patdb2", status READY, has 2 handler(s) for this service...
    Handler(s):
      "DEDICATED" established:0 refused:0 state:ready
         REMOTE SERVER
         (ADDRESS=(PROTOCOL=TCPS)(HOST=10.160.121.28)(PORT=1523))
      "DEDICATED" established:0 refused:0 state:ready
         REMOTE SERVER
         (ADDRESS=(PROTOCOL=TCP)(HOST=10.160.121.28)(PORT=1521))
The command completed successfully
```

If you see something showing it is blocked e.g.

```
    Handler(s):
      "DEDICATED" established:0 refused:0 state:blocked
         REMOTE SERVER
         (ADDRESS=(PROTOCOL=TCPS)(HOST=10.160.123.40)(PORT=1523))
```

This indicates a potential issue with the configuration, or that the database needs restarted so that LREG (there 12c process which handles listener registration) can pick up the sqlnet.ora changes showing the location of the wallet file.

You should also check that the configuration of the listener.ora and sqlnet.ora at the $TNS_ADMIN location was done correctly, as well as ensuring that the cluster configuration for the database references the TNS_ADMIN environment variable, or that the sqlnet.ora under the $ORACLE_HOME/network/admin location has the wallet configuration.

Finally, validate that the local_listener parameter references both TCP and TCPS ports.

Testing TLS

If not already done, run the configuration validation process and correct anything that it highlights as an FAIL:

```
dell_oracle_tls_auto.sh -status
```

Test Connectivity

Most of the configuration required to test connectivity has now been applied to the database ORACLE_HOME. The one thing which is missing is the TNS alias used to connect.

Start with a simple test with an invalid username and password:

```
sqlplus 1/1@'(DESCRIPTION = (ADDRESS = (PROTOCOL = TCPS)(HOST =
DBENGTS4DBSCN.us.dell.com)(PORT = 1523)) (CONNECT_DATA = (SERVER=DEDICATED)
(SERVICE_NAME=patdb.dev.amer.dell.com))(SECURITY=(SSL_SERVER_CERT_DN="CN=oracledb.dev
.amer.dell.com,OU=IEO,O=Dell Technologies")))'
```

If that is successful, you will see a response like:

```
SQL*Plus: Release 12.2.0.1.0 Production on Wed May 16 15:10:10 2018

Copyright (c) 1982, 2016, Oracle.  All rights reserved.

ERROR:
ORA-01017: invalid username/password; logon denied


Enter user-name:
```

If it fails, you might see a message like:

```
SQL*Plus: Release 12.2.0.1.0 Production on Wed May 16 21:02:26 2018

Copyright (c) 1982, 2016, Oracle.  All rights reserved.

ERROR:
ORA-12520: TNS:listener could not find available handler for requested type of
server
```

This is usually caused when the listener handlers are in status blocked, so review the configuration to ensure all the instructions were followed.

Now do a proper connectivity test using a TNS alias. Navigate to the database $ORACLE_HOME/network/admin folder, and copy any existing TNS alias for the database, and use it to create a new TCPS connection string. e.g. if this was the existing alias:

```
PATDB=
  (DESCRIPTION=
    (ADDRESS=
      (PROTOCOL=TCP)
      (HOST=DBENGTS4DBSCN.us.dell.com)
      (PORT=1521)
    )
    (CONNECT_DATA=
      (SERVER=dedicated)
      (SERVICE_NAME=patdb.dev.amer.dell.com)
```

```
    )
  )
```

Then create a new one like the following:

```
PATDB_TLS=
  (DESCRIPTION=
    (ADDRESS=
      (PROTOCOL=TCPS)
      (HOST=DBENGTS4DBSCN.us.dell.com)
      (PORT=1523)
    )
    (CONNECT_DATA=
      (SERVER=dedicated)
      (SERVICE_NAME=patdb.dev.amer.dell.com)
    )
    (SECURITY=
      (SSL_SERVER_CERT_DN="CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies")
    )
  )
```

Note that the PROTOCOL and PORT entries are now referring to TCPS and port 1523 respectively. The additional SECURITY section is very important if you want to validate that the database server you are connecting to is really the correct database server:

```
    (SECURITY=
      (SSL_SERVER_CERT_DN="CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies")
    )
```

The SSL_SERVER_CERT_DN in red, must exactly match the DN that was used as part of the certificate creation process. If you are unsure what that is, issue an orapki command against the wallet:

```
$ orapki wallet display -wallet /u01/app/oracle/admin/tls_wallet -jsafe
Oracle PKI Tool : Version 12.1.0.2
Copyright (c) 2004, 2014, Oracle and/or its affiliates. All rights reserved.

Requested Certificates:
User Certificates:
Subject:        CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies
Trusted Certificates:
Subject:        CN=Dell Technologies Issuing CA 101,O=Dell Technologies,L=Round
Rock,ST=Texas,C=US
Subject:        CN=Dell Technologies Root Certificate Authority 2018,OU=Cybersecurity,O=Dell
Technologies,L=Round Rock,ST=Texas,C=US
```

Note that the Subject entry for the User Certificate here, must exactly match the SSL_SERVER_CERT_DN entry in the TNS alias.

This DN name checking only occurs if you have the following entry in the database or client $ORACLE_HOME/network/admin/sqlnet.ora file:

```
SSL_SERVER_DN_MATCH=ON
```

Now that the entry is in the tnsnames.ora, run a tns ping to validate it:

```
$ tnsping PATDB_TLS

TNS Ping Utility for Linux: Version 12.1.0.2.0 - Production on 26-MAY-2017 05:17:51
```

```
Copyright (c) 1997, 2014, Oracle.  All rights reserved.

Used parameter files:
/u01/app/grid/product/12.1.0.2/grid_1/network/admin/sqlnet.ora


Used TNSNAMES adapter to resolve the alias
Attempting to contact (DESCRIPTION = (ADDRESS = (PROTOCOL = TCPS)(HOST =
DBENGTS4DBSCN.us.dell.com)(PORT = 1523)) (CONNECT_DATA = (SERVER = DEDICATED)
(SERVICE_NAME = patdb.dev.amer.dell.com)))
OK (80 msec)
```

Now do a test with SQLPlus:

```
$ sqlplus ora_readonly@PATDB_TLS

SQL*Plus: Release 12.1.0.2.0 Production on Fri May 26 07:30:43 2017

Copyright (c) 1982, 2014, Oracle.  All rights reserved.

Enter password:
Last Successful login time: Fri May 26 2017 07:29:30 -05:00

Connected to:
Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 - 64bit Production
With the Partitioning, Real Application Clusters, Automatic Storage Management, OLAP,
Advanced Analytics and Real Application Testing options

PATDB_TLS SQL> exit
Disconnected from Oracle Database 12c Enterprise Edition Release 12.1.0.2.0 - 64bit
Production
With the Partitioning, Real Application Clusters, Automatic Storage Management, OLAP,
Advanced Analytics and Real Application Testing options
```

Do a final test using openssl against both the VIP (local listener) and scan name. This will confirm that the certificate chain matches what is in the wallet e.g.

```
$ openssl s_client -connect ausdlqtcdb01-vip.us.dell.com:1523
CONNECTED(00000003)
depth=2 C = US, ST = Texas, L = Round Rock, O = Dell Technologies, OU =
Cybersecurity, CN = Dell Technologies Root Certificate Authority 2018
verify return:1
depth=1 C = US, ST = Texas, L = Round Rock, O = Dell Technologies, CN = Dell
Technologies Issuing CA 101
verify return:1
depth=0 O = Dell Technologies, OU = IEO, CN = oracledb.dev.amer.dell.com
verify return:1
---
Certificate chain
 0 s:/O=Dell Technologies/OU=IEO/CN=oracledb.dev.amer.dell.com
   i:/C=US/ST=Texas/L=Round Rock/O=Dell Technologies/CN=Dell Technologies Issuing CA
101
 1 s:/C=US/ST=Texas/L=Round Rock/O=Dell Technologies/CN=Dell Technologies Issuing CA
101
   i:/C=US/ST=Texas/L=Round Rock/O=Dell Technologies/OU=Cybersecurity/CN=Dell
Technologies Root Certificate Authority 2018
```

```
 2 s:/C=US/ST=Texas/L=Round Rock/O=Dell Technologies/OU=Cybersecurity/CN=Dell
Technologies Root Certificate Authority 2018
   i:/C=US/ST=Texas/L=Round Rock/O=Dell Technologies/OU=Cybersecurity/CN=Dell
Technologies Root Certificate Authority 2018
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIFPDCCBCSgAwIBAgITYQAANi7Bbk/H2/wamAAAAAA2LjANBgkqhkiG9w0BAQsF
…      <<< THIS BIT REMOVED TO REDUCE SIZE OF OUTPUT >>>
…
1g==
-----END CERTIFICATE-----
subject=/O=Dell Technologies/OU=IEO/CN=oracledb.dev.amer.dell.com
issuer=/C=US/ST=Texas/L=Round Rock/O=Dell Technologies/CN=Dell Technologies Issuing
CA 101
---
No client certificate CA names sent
Peer signing digest: SHA384
Server Temp Key: ECDH, P-256, 256 bits
---
SSL handshake has read 4248 bytes and written 415 bytes
---
New, TLSv1/SSLv3, Cipher is ECDHE-RSA-AES256-GCM-SHA384
Server public key is 2048 bit
Secure Renegotiation IS supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
SSL-Session:
    Protocol  : TLSv1.2
    Cipher    : ECDHE-RSA-AES256-GCM-SHA384
    Session-ID: 5DB9582E0A5755B825B62631EB944A0A8E25B8DE447EE466DD13BA3BAA60E844
    Session-ID-ctx:
    Master-Key:
16F5D0C76C70E10C877B7C1DBA94DF9A6B5F8D325AA8020B2B5230C89D4997D1752C1278CF4AD0789FE11
234641C2876
    Key-Arg   : None
    Krb5 Principal: None
    PSK identity: None
    PSK identity hint: None
    Start Time: 1592405581
    Timeout   : 300 (sec)
    Verify return code: 0 (ok)
---
```

Client/Application Server Configuration

Client Compatibility with TLS 1.2

The following table lists the compatibility of different versions of Oracle clients:

Oracle Client	TLS 1.2 support
Oracle Client 11.2.0.3 or earlier	None. Would need upgraded to 11.2.0.4 plus additional patches.
Oracle Client 11.2.0.4	Requires MES bundle patches or recent PSU. See note 2274242.1
Oracle Client 12.1.0.2	Supports TLS 1.2, but doesn't support the Dell approved cipher suites. If the Oracle Client is the Oracle Database, as long as it is patched with PSU January 2019 or later you are covered. If this is a client install then it will need the PSU January 2019 applied.
Oracle Client 12.2.0.1 and above	Supported.
JDBC 11.2.0.3 or earlier	No support for TLS 1.2.
JDBC 11.2.0.4	Apply patch 29672453. See note 2546551.1
JDBC 12.1.0.2	Apply patch or download latest 12.1.0.2 JJDBC driver from here. Note that to enable 1.2 need to include -Doracle.net.ssl_version='1.2' setting in the JVM start command. See notes 2367258.1 and 2436911.1 for more info.
JDBC 12.2.0.1 and above	Supported.
ODP.Net Unmanaged Driver	Support as per the Oracle Client version above
ODP.Net Managed Driver 12.1.0.2 and earlier	No support for TLS 1.2 or Distinguished name. Only TLS 1.0 and SSLv3 are supported which don't meet Dell's security standards.
ODP.Net Managed Driver 12.2.0.1	Supported. EZ connect syntax for TCPS only supported from 19.3 onwards.
ODP.Net Core 19.3 and above	Supported

Root Certificates and Client Wallets

The certificates issued within Dell have one of 2 root certificate authorities, one from before May 2, 2019 and one from after. As a result, the files for these 2 root certificates have been staged in the following location:

http://images-http.us.dell.com/images/oracle/tools/wallet/client/

The root directory contains the root certificates which can be imported into say a Java keystore, and the sub-directories contain Oracle Wallets which already have these 2 root certificates imported:

Index of /images/oracle/tools/wallet/client

Name	Last modified	Size	Description
Parent Directory		-	
12.1.0/	21-May-2019 04:48	-	
12.2.0/	14-May-2019 06:12	-	
19.0.0/	15-May-2019 05:26	-	
dell_root_ca.cer	09-May-2019 08:42	758	
dell_technologies_root.cer	09-May-2019 08:40	1.5K	

Windows PC/Server Wallet and sqlnet.ora Configuration

This assumes that there is already an Oracle 12.1/12.2/19.0 client installed on the PC, in a directory like C:\app\oracle\product\<<dbversion>>\client_1.

Configure TNS_ADMIN Environment

For the sake of this document it assumes it is in the following directory (referred to as ORACLE_HOME), but amend it based on the Oracle client version and location:

```
C:\app\oracle\product\12.1.0\client_1
```

The first step is to create a new wallet directory on windows under the %ORACLE_HOME%/network/admin (there might already be an environment variable called %TNS_ADMIN% pointing to this location):

```
C:\app\oracle\product\12.1.0\client_1\network\admin>mkdir wallet
```

Now you need to create/amend the sqlnet.ora file under the ORACLE_HOME/network/admin folder, by adding the wallet information to it (if the file doesn't exist then create it):

```
# Disable SSL client authentication
SSL_CLIENT_AUTHENTICATION = FALSE

# Check server DN to ensure it is correct
SSL_SERVER_DN_MATCH=ON

WALLET_LOCATION =
  (SOURCE =
    (METHOD = FILE)
    (METHOD_DATA =
      (DIRECTORY = C:\app\oracle\product\12.1.0\client_1\network\admin\wallet)
```

```
    )
  ).
```

Ensure that the parameter SSL_CLIENT_AUTHENTICATION is not in the sqlnet.ora file. If it is, it must be set to FALSE.

In addition, create a TNS alias in the ORACLE_HOME/network/admin/tnsnames.ora file. This should be the same alias created to test connectivity on the database server:

```
PATDB_TLS=
  (DESCRIPTION=
    (ADDRESS=
      (PROTOCOL=TCPS)
      (HOST=DBENGTS4DBSCN.us.dell.com)
      (PORT=1523)
    )
    (CONNECT_DATA=
      (SERVER=dedicated)
      (SERVICE_NAME=patdb.dev.amer.dell.com)
    )
    (SECURITY=
      (SSL_SERVER_CERT_DN="CN=DELL,OU=IEO,O=DB,L=RR,ST=TX,C=US")
    )
  )
```

Ensure the SSL_SERVER_CERT_DN value matches the DN of the certificate in the server wallet.

Use Pre-Created Oracle Wallet Files

If the version of the client is one of 12.1, 12.2 or 19c, and the root certificates are the same as the standard Dell Technologies root certificates, then copy the cwallet.sso and ewalllet.p12 files from one of these locations to the ORACLE_HOME/network/admin/wallet location, depending on the Oracle client version:

http://images-http.us.dell.com/images/oracle/tools/wallet/client/12.1.0/

http://images-http.us.dell.com/images/oracle/tools/wallet/client/12.2.0/

http://images-http.us.dell.com/images/oracle/tools/wallet/client/19.0.0/

You can use the Windows Command Prompt window (assuming it is configured to include the ORACLE_HOME/bin on the windows path) to view the wallet:

```
Command Prompt
C:\app\oracle\product\12.1.0\client_1\network\admin>
C:\app\oracle\product\12.1.0\client_1\network\admin>orapki wallet display -wallet wallet -jsafe
Oracle PKI Tool : Version 12.1.0.2
Copyright (c) 2004, 2014, Oracle and/or its affiliates. All rights reserved.

Requested Certificates:
User Certificates:
Trusted Certificates:
Subject:        CN=Dell Root Certificate Authority 2015,OU=MS PKI,O=Dell Inc.
C:\app\oracle\product\12.1.0\client_1\network\admin>
```

Not that this wallet just contains the Dell Root certificate, as only this certificate is need for client applications.

Create New Wallet Files Manually

If you wish to create your own wallet using the client, then obtain the Dell Root certificate in PEM format (which was generated as part importing the certificate into the server wallet) and copy it to the wallet directory. These certificates can also be downloaded from here.

Then open the Command Prompt and in the wallet directory run the following command

```
C:\app\oracle\product\12.1.0\client_1\network\admin>orapki wallet create -wallet
wallet -auto_login -pwd welcome1
Oracle PKI Tool : Version 12.1.0.2
Copyright (c) 2004, 2014, Oracle and/or its affiliates. All rights reserved.
```

Then import the root certificate using the following command:

```
C:\app\oracle\product\12.1.0\client_1\network\admin>orapki wallet add -wallet  wallet
-cert wallet/dellroot2015.cer -trusted_cert -pwd welcome1 -jsafe
Oracle PKI Tool : Version 12.1.0.2
Copyright (c) 2004, 2014, Oracle and/or its affiliates. All rights reserved.
```

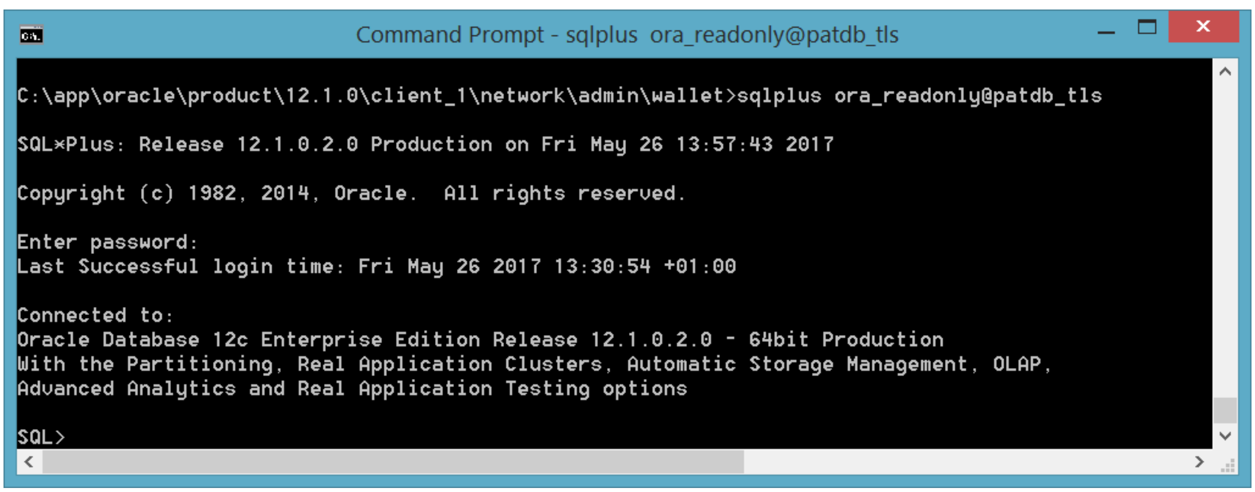
Test Client Connectivity

Run the command to check the certificate is in the wallet:

```
C:\app\oracle\product\12.1.0\client_1\network\admin>orapki wallet display -wallet
wallet -jsafe
Oracle PKI Tool : Version 12.1.0.2
Copyright (c) 2004, 2014, Oracle and/or its affiliates. All rights reserved.

Requested Certificates:
User Certificates:
Trusted Certificates:
Subject:        CN=Dell Root Certificate Authority 2015,OU=MS PKI,O=Dell Inc.
```

Then test connectivity using this alias from your client tools e.g. SQLPlus, TOAD:



If you hit an "ORA-28860: Fatal SSL error" error then it's likely that the ssl_client_authentication setting is wrong somewhere. Ensure that SSL_CLIENT_AUTHENTICATION is set to FALSE in listener.ora and sqlnet.ora files on the server, as well as the client sqlnet.ora. Further details are in this note:

SSL/TCPS Connection Is Failing With ORA-28860 (Doc ID 843500.1)

Oracle Data Provider for .NET (ODP.Net)

There are 3 different ODP.Net options for .Net/ADO.Net/C# based applications:

- ODP.Net Unmanaged Driver. This relies on the underlying Oracle client install for connectivity from .NET Framework applications. This is only available on Windows.
- ODP.Net Managed Driver. This is a standalone DLL which is packaged with .NET Framework applications. It requires no Oracle client to be installed, as the DLL is packaged with the application. This is only available on Windows.
- ODP.Net Core Managed Driver. This is a standalone DLL which is packaged with .NET Core applications. It requires no Oracle client to be installed, as the DLL is packaged with the application. .NET Core applications are platform independent and typically used for microservices (e.g. in PCF), or Docker. It can be used on Windows or Linux.

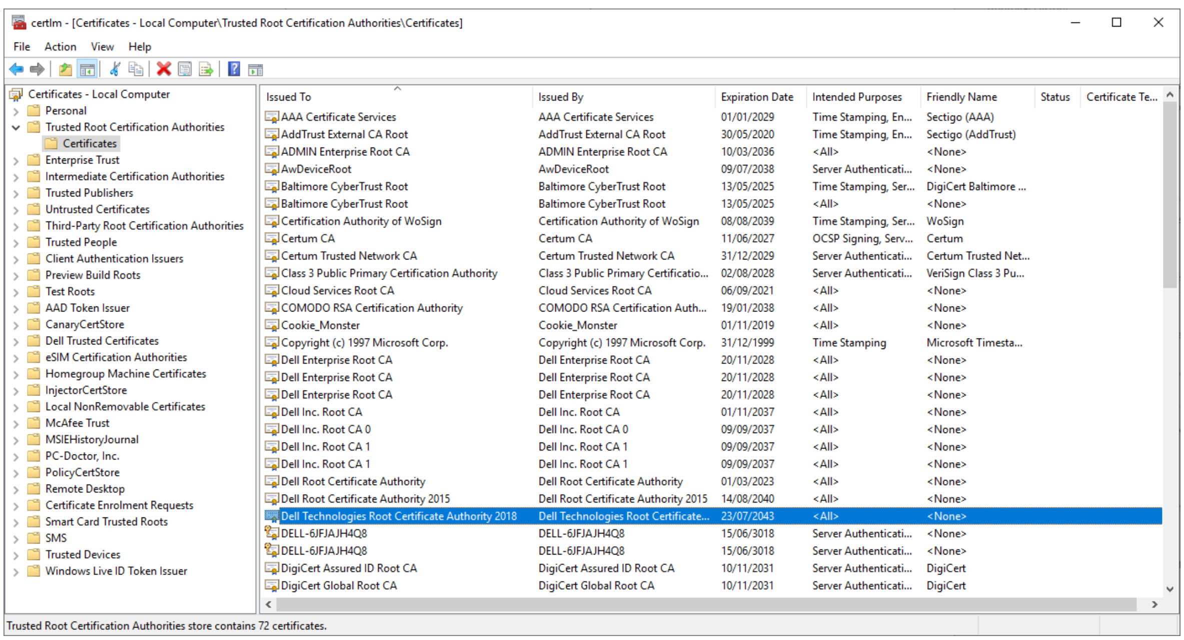
More information can be found here on the differences between .NET Core and .NET Framework. Generally .NET Framework is still in use for applications still deployed on Windows application servers. As applications migrate to PCF, they are being converted to use .NET Core.

More information on ODP.Net can be found here, and the documentation for the different releases can be found here.

ODP.Net Unmanaged Driver has additional functionality compared to the managed driver such as application continuity (available from 12.2+), so if there is a requirement for the application to experience zero outage if a RAC node fails, then consider using the unmanaged driver (Windows only i.e. not available in .Net core on Linux).

Installing the Dell Root Certificate in the Microsoft Certificate Store – Windows Only

The Dell root certificate should be installed by default on all Windows servers and laptops. Check this by opening the Control Panel and type ‘certificates’ in the search field. There should be an option to ‘Manage Computer Certificates’. Select that to open the application and navigate to ‘Trusted Root Certification Authorities’->Certificates. Check that the list includes the “Dell Technologies Root Certificate Authority 2018”:



If it doesn’t exist download the Dell Technologies Root CA 2018 from here. Once downloaded, open the folder and right click on the file and choose the option ‘Install Certificate’. The Certificate Import Wizard should open; select the option Local Machine and then continue to import the certificate into MCS.

ODP.Net Unmanaged Driver

As this relies on the Oracle client, follow the instructions in section “Windows PC/Server Wallet and sqlnet.ora Configuration”. Ensure the client is at least 12.1.0.2, for TLS 1.2 support.

ODP.Net Managed Driver

This has no dependency on the Oracle client (though sometimes it can be useful to have the Oracle client installed for troubleshooting using sqlplus).

ODP.Net requirements to support TCPS connections are similar to the Oracle Client requirements i.e.

1. The TNS connection strings should include TCPS, port 1523, and a “SECURITY=(SSL_SERVER_CERT_DN=” section which references the Distinguished Name (DN) for the database you are connecting to.
2. The following sqlnet.ora parameters:
    a. SSL_CLIENT_AUTHENTICATION = FALSE
    b. SSL_SERVER_DN_MATCH = ON
    c. SSL_VERSION=1.2
    d. WALLET_LOCATION … set to point to FILE or MCS.
3. A location for the Dell root trusted certificates This can be either of these:
    a. Oracle client wallet (cwallet.sso + ewallet.p12). This is used if the WALLET_LOCATION points to a file.
    b. Microsoft Certificate Services/MCS (assuming the Dell root certificates are already installed on the server). This is used if the WALLET_LOCATION points to MCS.

Some or all the sqlnet.ora parameters can be set using configuration options under ODP.Net, or can be set in the sqlnet.ora file e.g. the web.config could have the following set:

```
<setting name="WALLET_LOCATION"
value="(SOURCE=(METHOD=FILE)(METHOD_DATA=(DIRECTORY=c:\temp\wallet)))"/>
<setting name="SSL_VERSION" value= "1.2" />
<setting name="SSL_CLIENT_AUTHENTICATION" value = "FALSE" />
<setting name="SSL_SERVER_DN_MATCH" value= "ON" />
```

If MCS is in use the WALLET_LOCATION would be set as:

```
<setting name="WALLET_LOCATION" value="(SOURCE=(METHOD=MCS))"/>
```

It is recommended to use MCS on Windows servers, as this avoids the need to package up the Oracle wallet files with the .Net build.

Sometimes the MCS option doesn’t work, so specify the wallet location using the ‘METHOD=FILE’ option and download the cwallet.sso and ewallet.p12 files from here. There are sub-directories for each version of the Oracle client, so download the files relevant to the version of the client in use.

An alternative to setting these in the config files, is just to include a sqlnet.ora file in the base direct of the application (or in the location specified by the TNS_ADMIN variable if that is set). The file would contain the following:

```
# Disable SSL client authentication
SSL_CLIENT_AUTHENTICATION = FALSE
# Check server DN to ensure it is correct
SSL_SERVER_DN_MATCH=ON
# Force TLS v1.2
SSL_VERSION=1.2
# Set the wallet location
WALLET_LOCATION=(SOURCE=(METHOD=MCS))
```

If OUD is being used for TNS name lookup, there might also be an entry like the following:

```
NAMES.DIRECTORY_PATH=(TNSNAMES, LDAP, EZCONNECT)
```

And an associated ldap.ora with connection details for the OUD directory.

Once the setup in complete, it should just be a case of specifying the Oracle connection string. If connection strings are in a tnsnames.ora file they will take the format:

```
DBUP2D=(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=zzzdlolalandb01.us.dell.com)(PORT=1
523))(CONNECT_DATA=(SERVER=DEDICATED)(SERVICE_NAME=dbup2d.dev.amer.dell.com))(SECURIT
Y=(SSL_SERVER_CERT_DN="CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies")))
```

Note that the 'DN' is in double quotes, and the DN should match the server DN name.

If the connection string is in a .config file or directly in the code, the whole connection string will be surrounded by double quotes. If that is the case the DN should be in single quotes e.g.

```
"(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=zzzdlolalandb01.us.dell.com)(PORT=1523))(
CONNECT_DATA=(SERVER=DEDICATED)(SERVICE_NAME=dbup2d.dev.amer.dell.com))(SECURITY=(SSL
_SERVER_CERT_DN='CN = oracledb.dev.amer.dell.com, OU = IEO, O = Dell
Technologies')))"
```

ODP.Net CORE Managed Driver

This can be downloaded from nuget here. It is recommended to use the latest version.

This has no dependency on the Oracle client but requires a slightly different setup to the ODP.Net Managed Driver, as ODP.Net CORE does not support Oracle configuration files like web.config. Instead it uses the .NET Configuration API in lieu of a configuration file. See here for more info on the different parameters.

Note that the ODP.Net CORE managed driver supports a subset of the features of the ODP.Net managed driver, and sometimes the method of configuring connection properties is different e.g. ODP.Net CORE does not support OUD lookup of TNS connection strings.

Rather than using the configuration API in code, it is recommended to package a sqlnet.ora file with the application, as it is easier to amend a configuration file, rather than to amend code and recompile. ODP.Net core will look for the sqlnet.ora (and tnsnames.ora) file in the following precedence order:

1. Directory set in OracleConfiguration.TnsAdmin property
2. Directory of the running ODP.NET Core assembly
3. Current working directory

This would be the configuration file for Windows based .NET Core applications:

```
# Disable SSL client authentication
SSL_CLIENT_AUTHENTICATION = FALSE
# Check server DN to ensure it is correct
SSL_SERVER_DN_MATCH=ON
# Force TLS v1.2
SSL_VERSION=1.2
# Set the wallet location
WALLET_LOCATION=(SOURCE=(METHOD=MCS))
```

Unfortunately, the Linux Diego cells on PCF do not currently support MCS (they are based on the Ubuntu Linux OS), so the wallet location must refer to a FILE/DIRECTORY. The resulting sqlnet.ora file would look like the following:

```
# Disable SSL client authentication
SSL_CLIENT_AUTHENTICATION = FALSE
# Check server DN to ensure it is correct
SSL_SERVER_DN_MATCH=ON
# Force TLS v1.2
```

```
SSL_VERSION=1.2
# Set the wallet location
WALLET_LOCATION=(SOURCE=(METHOD=FILE)(METHOD_DATA=(DIRECTORY=.)))
```

The dot '.' represents the current working directory, which should be the same as the location of the sqlnet.ora. That means it will look for the files cwallet.sso and ewallet.p12 in the same location.

Download the cwallet.sso and ewallet.p12 files from here. There are sub-directories for each version of the Oracle client, so download the files relevant to the version of the client in use (e.g. ODP.Net Core 2.19.70 would require the 19c wallet files).

If you don't want to package the wallet files in the base directory. Create a subdirectory called tls_wallet, put the 2 wallet files in there and amend the WALLET_LOCATION as follows:

```
WALLET_LOCATION=(SOURCE=(METHOD=FILE)(METHOD_DATA=(DIRECTORY=./tls_wallet)))
```

Note: Use Linux forward slash (/) rather than Windows back slash (\) to delimit the directories as these work on both Windows and Linux.

For PCF it's important to use relative paths like this, because there is no control over the location of the published files.

When building/publishing the application, be sure to set the properties on the sqlnet.ora, cwallet.sso and ewallet.p12 to 'Copy if newer'. This will ensure they get included automatically. The csprof file should contain something like the following:

```
  <ItemGroup>
    <None Update="sqlnet.ora">
      <CopyToOutputDirectory>PreserveNewest</CopyToOutputDirectory>
    </None>
    <None Update="tls_wallet\cwallet.sso">
      <CopyToOutputDirectory>PreserveNewest</CopyToOutputDirectory>
    </None>
    <None Update="tls_wallet\ewallet.p12">
      <CopyToOutputDirectory>PreserveNewest</CopyToOutputDirectory>
    </None>
    <None Update="tnsnames.ora">
      <CopyToOutputDirectory>PreserveNewest</CopyToOutputDirectory>
    </None>
  </ItemGroup>
```

Finally the TNS connect string will follow the same rules and format as the standard ODP.Net Managed Driver and can either be in a tnsnames.ora file, or directly in the code using something like:

```
OracleConfiguration.OracleDataSources.Add("DBUP2D",
"(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=zzzdlolalandb01.us.dell.com)(PORT=1523))(
CONNECT_DATA=(SERVER=DEDICATED)(SERVICE_NAME=dbup2d.dev.amer.dell.com))(SECURITY=(SSL
_SERVER_CERT_DN='CN=oracledb.dev.amer.dell.com,OU=IEO,O=Dell Technologies')))");
```

Oracle JDBC Connectivity

Oracle JDBC is used for any Java applications to connect to Oracle databases. Typical applications using Java in Dell:

1. Oracle Weblogic for J2EE based applications, or 3rd party applications which use Oracle Weblogic.
2. PCF Java Build Pack e.g. Grails, Groovy, Java Main, Play Framework, Spring Boot, and Servlet.

Regardless of which of these mechanisms is used, they both need a Java trust store, which is basically a Java keystore which contains trusted root certificates. The Dell root certificate isn't in the standard Java cacerts trusted keystore so it is necessary to add it.

Once that is done, there will then be additional configuration depending on the Java platform in use. Note 2436911.1 has additional information for using JDBC with TLS 1.2.

Creating a Trusted KeyStore

Note: In the following section the commands assume this is Linux, but for Windows replace references to Linux environment variables e.g. $JRE_HOME by their Windows equivalent e.g. %JRE_HOME%. Replace Linux forward slash (/) directory separators by Windows back slash (\) based separators.

Start by downloading the Dell root certificates from here, and copy them to the location where the Java runtime environment. If this is Linux it can be done directly on the e.g. make a directory called /dbscratch/certificates and use wget to download the file and unzip it:

```
mkdir -p /dbscratch/certificates
cd /dbscratch/certificates
wget http://pki.dell.com/java/dellcas.zip
unzip dellcas.zip
ls -l
```

The output should be:

```
total 24
-rw-r--r-- 1 oracle oinstall 3922 Jul  8  2019 dellcas2015.jks
-rw-r--r-- 1 oracle oinstall 6598 Jun 19  2019 dellcas2018.jks
-rw-r--r-- 1 oracle oinstall 5632 Jul  8  2019 dellcas.zip
-rw-r--r-- 1 oracle oinstall  482 Jul  8  2019 README.txt
```

If you 'cat' the README.txt it shows the instructions to import the keystore:

```
To import the 2018 trust chain into your cacerts:

keytool -importkeystore -srckeystore dellcas2018.jks -destkeystore
(your_keystore_or_cacerts_file) -srcstorepass dell2018

To import the 2015 trust chain into your cacerts:

keytool -importkeystore -srckeystore dellcas2015.jks -destkeystore
(your_keystore_or_cacerts_file) -srcstorepass dell2015

You will be prompted for the password on your keystore. If you haven't set this
password, the default is usually "changeit".
```

Now cocate the Java runtime executables used for your application e.g. on an Oracle Database server it will be under $ORACLE_HOME/jdk/jre/bin

```
ls -l $ORACLE_HOME/jdk/jre/bin
```

```
total 460
```

```
-rwxr-xr-x 1 oracle oinstall   6264 Mar 19 12:04 ControlPanel
-rwxr-xr-x 1 oracle oinstall   8712 Mar 19 12:04 java
-rwxr-xr-x 1 oracle oinstall 140488 Mar 19 12:04 javaws
-rwxr-xr-x 1 oracle oinstall   6264 Mar 19 12:04 jcontrol
-rwxr-xr-x 1 oracle oinstall   8832 Mar 19 12:04 jjs
-rwxr-xr-x 1 oracle oinstall   8832 Mar 19 12:04 keytool
-rwxr-xr-x 1 oracle oinstall   8904 Mar 19 12:04 orbd
-rwxr-xr-x 1 oracle oinstall   8832 Mar 19 12:04 pack200
-rwxr-xr-x 1 oracle oinstall   8840 Mar 19 12:04 policytool
-rwxr-xr-x 1 oracle oinstall   8832 Mar 19 12:04 rmid
-rwxr-xr-x 1 oracle oinstall   8840 Mar 19 12:04 rmiregistry
-rwxr-xr-x 1 oracle oinstall   8840 Mar 19 12:04 servertool
-rwxr-xr-x 1 oracle oinstall   8904 Mar 19 12:04 tnameserv
-rwxr-xr-x 1 oracle oinstall 186960 Mar 19 12:04 unpack200
```

Set an environment variable to the JRE home directory e.g.

```
export JRE_HOME=$ORACLE_HOME/jdk/jre
```

The existing trusted certificate file can be found under $JRE_HOME/lib/security/cacerts:

```
ls -l $JRE_HOME/lib/security/
total 164
-rw-r--r-- 1 oracle oinstall   4054 Mar 19 12:04 blacklist
-rw-r--r-- 1 oracle oinstall   1273 Mar 19 12:04 blacklisted.certs
-rw-r--r-- 1 oracle oinstall 105352 Mar 19 12:04 cacerts
-rw-r--r-- 1 oracle oinstall   2466 Mar 19 12:04 java.policy
-rw-r--r-- 1 oracle oinstall  45027 Mar 19 12:04 java.security
-rw-r--r-- 1 oracle oinstall     98 Mar 19 12:04 javaws.policy
drwxr-xr-x 4 oracle oinstall     38 Apr 29 08:52 policy
-rw-r--r-- 1 oracle oinstall      0 Mar 19 12:04 trusted.libraries
```

It isn't recommended to add the Dell root certificates to this file, as if Java is upgraded then this file will be overwritten. Instead specify a location which will remain untouched during any Java upgrade. Copy the existing cacerts file to this new location, and then add the Dell root certificate e.g.

```
mkdir -p
cp $JRE_HOME/lib/security/cacerts /dbscratch/certificates/wallet
```

Then add the Dell technologies root certificates to the file:

```
$JRE_HOME/bin/keytool -importkeystore -srckeystore
/dbscratch/certificates/dellcas2018.jks -destkeystore
/dbscratch/certificates/wallet/cacerts -srcstorepass dell2018
$JRE_HOME/bin/keytool -importkeystore -srckeystore
/dbscratch/certificates/dellcas2015.jks -destkeystore
/dbscratch/certificates/wallet/cacerts -srcstorepass dell2015
```

Enter the default keystore password of 'changeit', confirm it then it should show the certificates as imported:

```
$ $JRE_HOME/bin/keytool -importkeystore -srckeystore
/dbscratch/certificates/dellcas2018.jks -destkeystore
/dbscratch/certificates/wallet/cacerts -srcstorepass dell2018
Importing keystore /dbscratch/certificates/dellcas2018.jks to
/dbscratch/certificates/wallet/cacerts...
Enter destination keystore password:
Re-enter new password:
Entry for alias dell2018iss102 successfully imported.
```

```
Entry for alias dell2018iss101 successfully imported.
Entry for alias dell2018iss302 successfully imported.
Entry for alias dell2018iss301 successfully imported.
Entry for alias dellca2018 successfully imported.
Import command completed:  5 entries successfully imported, 0 entries failed or
cancelled
[oracle@zzzdlolalandb01.us.dell.com /dbscratch/certificates/wallet]
$ $JRE_HOME/bin/keytool -importkeystore -srckeystore
/dbscratch/certificates/dellcas2015.jks -destkeystore
/dbscratch/certificates/wallet/cacerts -srcstorepass dell2015
Importing keystore /dbscratch/certificates/dellcas2015.jks to
/dbscratch/certificates/wallet/cacerts...
Enter destination keystore password:
Entry for alias dellca2015 successfully imported.
Entry for alias dell2015iss102 successfully imported.
Entry for alias dell2015iss101 successfully imported.
Entry for alias dell2015iss302 successfully imported.
Entry for alias dell2015iss301 successfully imported.
Import command completed:  5 entries successfully imported, 0 entries failed or
cancelled
```

This wallet can then be used in the subsequent sections.

Configuring Weblogic to use Trusted Key Store

Weblogic must be running at least JRE/JDK 7 or above. It is also recommended to patch Weblogic with the latest OJDBC drivers, following the instructions for your Weblogic release here:

How To Update the JDBC and UCP Drivers Bundled with WebLogic Server 10.3.6 and 12c (Doc ID 1970437.1)

This will ensure that TLS 1.2 works 100%, as older OJDBC drivers (12.1.0.2) lack TLS 1.2 support without additional patches. The later drivers (12.2+) also have support for the TNS parameter TRANSPORT_CONNECT_TIMEOUT, which is very important for class 1 systems, to protect against connect timeouts if the primary data centre goes offline.

If the database is running 19c, then it's highly recommended that Weblogic is upgraded to 12.2.1.4, which has native support for the 19c JDBC drivers. If a Weblogic upgrade isn't feasible, consider downloading and installing the latest JDBC drivers, following the guidance in the following note for configuring them:

https://blogs.oracle.com/weblogicserver/oracle-database-122-feature-support-with-weblogic-server

The Oracle Fusion Middleware Administration Guide documents the steps required to configure the Weblogic JDBC connection to allow it to connect using TCPS:

https://docs.oracle.com/middleware/1221/core/ASADM/sslconfig.htm#ASADM10351

Section 6.6.1.2 "SSL-Enabling a Data Source" has the steps to do this. The high level steps are:

1. Create a truststore and add the root certificate (which is created when SSL-enabling the database) as a trusted certificate to the truststore.
2. In the Oracle WebLogic Server Administration Console, navigate to the Connection pool tab of the data source that you are using, and update the JDBC properties box.
3. Update the connection URL to use TCPS and port 1523, and the additional "SECURITY=(SSL_SERVER_CERT_DN…" attributes.
4. Test the connection.

For step 1, that should have been done in the previous section "Creating a Trusted KeyStore".

To view the contents of the trust store, you can run the following command:

```
$JRE_HOME/bin/keytool -list -keystore /dbscratch/certificates/wallet/cacerts -v -
storepass changeit
```

If you are using an older version of Java (i.e. Java 6), and you see an error like the following:

```
$JRE_HOME/bin/keytool -keystore ./cacerts -storepass changeit -importcert -
trustcacerts -alias DellRootCA2018 -file dell_technologies_root.cer
keytool error: java.security.NoSuchAlgorithmException: SHA256withECDSA Signature not
available
```

This is because older versions of Java cannot handle SHA256withECDSA based certificates, which is what we use in Dell. A version of Weblogic running a later version of JDK/JRE 7 or 8 is recommended.

Now log onto the Weblogic Server Administration Console, navigate to the connection pool tab of the data source that is in use, and update the JDBC properties box with the following:

```
javax.net.ssl.trustStore /dbscratch/certificates/wallet/cacerts
javax.net.ssl.trustStoreType=JKS
javax.net.ssl.trustStorePassword=changeit
```

Ensure the trustStore location matches the location of your trusted certificate file, and the password is correct (if you have changed the default password).

Finally, amend the URL database connection string, to something like the following:

```
jdbc:oracle:thin:@(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=DBENGTS4DBSCN.us.dell.co
m)(PORT=1523))(CONNECT_DATA=(SERVER=dedicated)(SERVICE_NAME=patdb.dev.amer.dell.com))
(SECURITY=(SSL_SERVER_CERT_DN="CN=DELL,OU=IEO,O=DB,L=RR,ST=TX,C=US")))
```

Ensure the protocol (TCPS), hostname, port (1523) and service_name are set correctly for your database, and the SSL_SERVER_CERT_DN distinguished name (DN) exactly matches the DN of the server certificate in the database wallet.

Recycle the connection pool and test the connection.

If the database has had the January 2019 PSU applied to it, connections might fail unless you force Weblogic to initiate a TLS 1.2 connection. This is because by default it will attempt an SSLv2 handshake, and the database from the January 2019 PSU will just terminate the connection, the moment it receives an SSLv2 handshake.

To force Weblogic to use TLS 1.2, set the property oracle.net.ssl_version=1.2. This can either be done as a system property during startup (-Doracle.net.ssl_version="1.2"), or in the datasource properties.

To force Weblogic to validate the DN, then set the property oracle.net.ssl_server_dn_match="true".

Configuring Java Application to use Trusted Key Store

Note. This assumes that there is a file /dbscratch/certificates/wallet/cacerts now containing the Dell root certificates.

If the Java application is not running under Weblogic (e.g. Spring Boot) then it will be necessary to download and install the latest Oracle JDBC driver, and set some runtime parameters to specify the location of the truststore. There is a very good article here that explains the settings that need to be added. The section ‘SSL connection using TLSv1.2 with JKS’ has the main steps.

Summarising the information from that article if a patched JDBC 12.1.0.2 driver is in use, then you need to include the following settings in the java environment. This can be done in code, or passed as parameters to the java startup script (the mechanism will depend on Java application runtime environment):

```
-Doracle.net.ssl_version="1.2"
-Doracle.net.ssl_server_dn_match="true"
-Djavax.net.ssl.trustStore="/dbscratch/certificates/wallet/cacerts"
-Djavax.net.ssl.trustStorePassword="changeit"
```

WARNING: The 12.1.0.2 driver really should not be used. As well as needing a patch to make it work with TLS 1.2, it also has no support for the TNS parameter TRANSPORT_CONNECT_TIMEOUT, which means connection delays if the primary data centre goes down.

If the 12.2.0.1 or later driver is in use, then set the following:

```
-Doracle.net.ssl_server_dn_match="true"
-Djavax.net.ssl.trustStore="/dbscratch/certificates/wallet/cacerts"
-Djavax.net.ssl.trustStorePassword="changeit"
```

Note that do not use the EZConnect format in the JDBC connection string. You must use a full connection string similar to the following:

```
jdbc:oracle:thin:@(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=DBENGTS4DBSCN.us.dell.co
m)(PORT=1523))(CONNECT_DATA=(SERVER=dedicated)(SERVICE_NAME=patdb.dev.amer.dell.com))
(SECURITY=(SSL_SERVER_CERT_DN="CN=DELL,OU=IEO,O=DB,L=RR,ST=TX,C=US")))
```

Otherwise some of the HA features of Oracle will fail to work in the event of a database note failure.

Oracle SQL Developer Configuration for TCPS Connectivity

Oracle SQL Developer (SQLD) does not support the thin JDBC driver for TCPS connectivity based on the advice in this note:

[Java error: 'Unable to find valid certification path to requested target' when trying to connect via TCPS (Doc ID 1457274.1)]()

This means in order to use SQLD to connect to a TCPS enabled database, you must follow the steps above and install/configure the Oracle client, ensuring the sqlnet.ora contains reference to the WALLET_LOCATION, and that the wallet contains the root certificates for the database you are connecting to.

Once that is in place, amend the SQLD preferences to allow it to use OCI:

Menu -> Tools -> Preferences -> Database -> Advanced

Click the ‘Use Oracle Client’ check box and enter the location of the ORACLE_HOME where the client is installed.

Click on ‘Use OCI/Thick driver’ check box and enter the location of the tnsnames directory, usually %ORACLE_HOME%\network\admin.

Click OK to exit the preferences. Now create a new connection (using the green + button) and specify the name/username and password as normal. There’s now 2 options available for the connection.

Choose Connection Type Custom JDBC and use a Custom JDBC URL of the format (note the “oci”):

jdbc:oracle:oci:@(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=<<hostname>>)(PORT=1523))(CONNECT_DATA=(SERVER=DEDICATED)(SERVICE_NAME=<<servicename>>))(SECURITY=(SSL_SERVER_CERT_DN="<<DN name>>")))

Alternatively choose Connection Type TNS and either choose an existing TNS alias in the Network Alias drop down, or go for Connect Identifier and enter a string of the format:

(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=<<hostname>>)(PORT=1523))(CONNECT_DATA=(SERVER=DEDICATED)(SERVICE_NAME=<<servicename>>))(SECURITY=(SSL_SERVER_CERT_DN="<<DN name>>")))

This is basically the same string as the JDBC connction but without the “jdbc:oracle:oci:@” prefix.

Removing TCP/1521 Configuration

Nexpose security scans highlight a "Database Open Access" vulnerability if the TCP/1521 configuration is still in place. While this vulnerability is technically a PCI only vulnerability, it highlights that the database is still breaking Dell's security policies that all data from restricted, highly restricted and PCI/KFS/Banking databases should be encrypted in transport.

This section lists the pre-requisites and steps for disabling TCP/1521, leaving just TCPS/1523 for all traffic.

Checklist for Removing TCP/1521

The following pre-requisites are required before disabling TCP/1521:

1. Ensure GRID, ASM and the database are correctly configured for TCPS/1523.
2. All applications and user/client connections to the database must be configured to use TCPS/1523.
3. For class 1's, ensure FSFO, Data Guard Broker and Data Guard/archive log shipping has been configured to use TCPS/1523.
4. Deploy OEM Agents which have support for TCPS/1523 i.e. agents which are bundled with a wallet configuration which includes the Dell root certificates.

Ensure Database is Configured for TCPS/1523

Use the dell_oracle_tls_auto.sh tool to check the status of the TLS configuration e.g.

```
dell_oracle_tls_auto.sh -status
```

All checks should pass as green. The main things to note are:

1. There is no sign of a blocked scan listener for the running database.
2. That the wallet permissions are correct (750) so that OEM can connect just using TCPS/1523.

Validate Connections are Using TCPS Protocols

There could be 100's of applications and servers connecting to the database servers, so it is challenging to validate all connections are using TCPS. Ensure that all non-production environments and databases have port TCP/1521 disabled. This should flush out many of the issues during testing.

For both test and production environments run the following script to extract the connection protocol information from the listener logs:

```
/misc/software/database/oracle/dba_scripts/user/lsnrlog_tcps_extract.sh
```

It should output a file which can be scanned e.g.

```
Processing
/u01/app/oracle/diag/tnslsnr/ausul2affndb01/listener/trace/listener.log_20200519_1305
.log
Processing /u01/app/oracle/diag/tnslsnr/ausul2affndb01/listener/trace/listener.log

Completed processing all the listener logs.....

/dbscratch/listener_testdata_ausul2affndb01.log contains required information from
the listener logs.

   Examine this file for anything still connecting using tcp e.g. grep -v tcps
/dbscratch/listener_testdata_ausul2affndb01.log .
```

You can then use a grep command to check for any entries not using tcps e.g.

```
grep -v tcps /dbscratch/listener_testdata_ausul2affndb01.log
Host=00DB5B8D-F65A-
4|Protocol=tcp|Program=C:\Users\vcap\app\.cloudfoundry\hwc.exe|ServiceName=affg2_12c_
default.sit.amer.dell.com|User=vcap
Host=049D8051-F2AD-
4|Protocol=tcp|Program=C:\Users\vcap\app\.cloudfoundry\hwc.exe|ServiceName=affg2_12c_
default.sit.amer.dell.com|User=vcap
Host=0b038fc7-bc43-40c5-5e14-
0591|Protocol=tcp|Program=/home/vcap/app/DELL.HT.LookUpServices.dll|ServiceName=affg2
_12c_default.sit.amer.dell.com|User=vcap
Host=0c24f1ca-b68f-4626-4d23-
dcb6|Protocol=tcp|Program=/home/vcap/app/DELL.HT.RabbitMQ.PTMExtractPub.dll|ServiceNa
me=affg2_12c_default.sit.amer.dell.com|User=vcap
Host=1D7A65B8-414F-
4|Protocol=tcp|Program=C:\Users\vcap\app\.cloudfoundry\hwc.exe|ServiceName=affg2_12c_
default.sit.amer.dell.com|User=vcap
Host=2069e0d9-f831-4bdf-5f83-
7dc7|Protocol=tcp|Program=/home/vcap/app/PartnerAccount.dll|ServiceName=affg2_12c_def
ault.sit.amer.dell.com|User=vcap
Host=221b8c62-934f-46ce-5947-
c6c1|Protocol=tcp|Program=/home/vcap/app/DELL.HT.RabbitMQ.PTMExtractPub.dll|ServiceNa
me=affg2_12c_default.sit.amer.dell.com|User=vcap
Host=2258aba5-a209-41af-4667-
ccd1|Protocol=tcp|Program=.cloudfoundry\hwc.exe|ServiceName=affg2_12c_default.sit.ame
r.dell.com|User=vcap
```

In the example above it shows lots of PCF based micro-services still using Protocol tcp.

Configure Data Guard/DG Broker/FSFO for TCPS

Follow the configuration guides for these tools to use TCPS.

Deploy OEM Agents with TCPS Support

OEM needs the following in order to monitor database servers configured purely with TCPS:

1. OMS and OM agent patched with middleware patch 20629366.
2. OMS and OM agent bundled with a PKCS#12 trusted certificate store, containing the Dell root certificates.
3. OMS and OM agent with the em.targetauth.db.pki.TrustStoreXXXX parameters set to use this trusted certificate store.

Confirm with the tools team whether the version of OEM in use for the database has been configured for TCPS/1523, and any additional steps required to configure the database and listener monitoring configuration.

Configure Database for TCPS/1523 only

Assuming all the pre-requisites are in place, all that is required to configure a database to only register with TCPS/1523 is:

1. The local_listener parameter should point to the local host/VIP using TCPS and 1523 only e.g.

```
(DESCRIPTION=(ADDRESS=(PROTOCOL=TCPS)(HOST=1.2.3.4)(PORT=1523)))
```

2. For RAC clusters, the remote_listener parameter must be changed from just pointing to scanname:1521 to point to each of the 3 scan IP's e.g.

```
(ADDRESS_LIST=(ADDRESS=(PROTOCOL=TCPS)(HOST=1.2.3.4)(PORT=1523))(ADDRESS=(PROTOCOL=TC
PS)(HOST=1.2.3.5)(PORT=1523))(ADDRESS=(PROTOCOL=TCPS)(HOST=1.2.3.6)(PORT=1523)))
```

The tls automation script can be used to do this configuration, using the following (change dbname for the database you wish to configure):

```
dell_oracle_tls_auto.sh -db dbname -tcps_only
```

Check the log and it will show if the parameters have been set:

```
[INFO]    local_listener parameter already configured for tcps/1523
[INFO]      TCPS only configuration so ensure only 1523/TCPS is configured in local_listener
[INFO]      Local_listener: (ADDRESS=(PROTOCOL=TCPS)(HOST=10.176.204.16)(PORT=1523))
[INFO]      Setting local_listener: alter system set local_listener="(ADDRESS=(PROTOCOL=TCPS)(HOST=10.176.204.16)(PORT=1523))" scope=both sid='dbup4d';
```

Do a final sanity check of the following:

1. Connect to the database and validate the local_listener and remote_listener (only for RAC) parameters.
2. Run lsnrctl status commands against each listener to validate the database services are registered.
3. Use the dell_oracle_tls_auto.sh -status command e.g.

```
dell_oracle_tls_auto.sh -status
```

Configure GI for TCPS/1523 only

The final step is to disable/remove the TCP/1521 configuration from GI, which removes it from the listener and scan listeners. It also sets the local_listener parameter for ASM to point to TCPS/1523. The tls automation script can be used for this:

```
dell_oracle_tls_auto.sh -grid -tcps_only
```

This will also correct any missing or incorrect parameters in the listener.ora and sqlnet.ora.

The -grid -tcps_only option will use srvctl modify listener commands to set the listener endpoints to TCPS:1521:

```
$GRID_HOME/bin/srvctl modify listener -p “TCPS:1523”
```

For RAC it will do the same for the scan listeners:

```
$GRID_HOME/bin/srvctl modify scan_listener -p “TCPS:1523”
```

Note that these commands will restart the listeners and so should be done as part of a CRQ or downtime on production.

For ASM the local_listener is set to the local host (Oracle Restart) or VIP (RAC), otherwise the ASM services won’t be available which will prevent OEM from monitoring ASM:

```
(ADDRESS=(PROTOCOL=TCPS)(HOST=1.2.4.4)(PORT=1523))
```

The SID is +ASM for Oracle Restart, and is +ASM1 to +ASMn for RAC.

Maintenance

OUD Naming Standard for TLS

Existing OUD connection aliases are of the format <DBNAME>_RW_OUD or <DBNAME>_RO_OUD. These names reflect the underlying database service name used to connect, so it wouldn't make sense to create dedicated services for TLS, as the existing services can be used.

With that in mind, any connection aliases for TLS should be onboarded to OUD with _TLS appended to the name e.g.

<DBNAME>_RW_OUD_TLS

The connection string for the _TLS alias should be very similar to the non _TLS alias, with PROTOCOL TCPS replacing TCP, PORT 1523 replacing 1521, and the additional security section:

(SECURITY=(SSL_SERVER_CERT_DN="CN=<<DN name>>,OU=IEO,O=Dell Technologies"))

Updating the Server Certificate on a Running Database

A server certificate is only valid for 2 years, and so every 2 years the database wallet will need updated with a new certificate.

Fortunately, the database wallet is only accessed when a new connection is made to the database, so ideally try to schedule the deployment of a new certificate during downtime. If that isn't possible though, the wallet can be replaced/updated while the database is running, and any new connections will use this wallet. The listeners will need to be bounced though in case they are caching the old certificate.

This assumes that the TLS configuration of the listener.ora and sqlnet.ora is already in place and working.

The steps would be:

1. Create a new wallet in another location, using the 'Certificate Request Process' section of this document.
    a. **Ensure the same distinguished name is used for the new certificate as on the existing database, otherwise every client will need to change the connect string.**
2. Backup the existing wallets in /u01/app/oracle/admin/tls_wallet.
3. Copy the new wallets to /u01/app/oracle/admin/tls_wallet, replacing the existing wallets.
4. Perform a rolling restart of the local listeners and scan listeners.

The Dell Root Certificate is valid until the 14th August 2040, so there should be no requirement for a long time to update any client wallets.

That said, as part of installing a new certificate, always check that the root certificate is unchanged. You can check the expiry dates of the certificate using openssl e.g.

```
$ openssl x509 -in DellRootCertificateAuthority2015.pem -noout -text
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            62:8b:f2:6a:05:7a:0f:88:49:26:94:a6:4b:0e:d4:1b
    Signature Algorithm: ecdsa-with-SHA256
        Issuer: O=Dell Inc., OU=MS PKI, CN=Dell Root Certificate Authority 2015
        Validity
            Not Before: Aug 14 21:08:44 2015 GMT
            Not After : Aug 14 21:18:42 2040 GMT
        Subject: O=Dell Inc., OU=MS PKI, CN=Dell Root Certificate Authority 2015
```

```
        Subject Public Key Info:
            Public Key Algorithm: id-ecPublicKey
                Public-Key: (256 bit)
                pub:
                    04:be:15:31:52:1a:6f:c8:4f:f4:f7:7c:fb:99:b5:
                    5c:ee:34:7c:07:69:52:5e:85:47:d4:b0:73:d8:7d:
                    fd:a3:69:71:9a:89:19:4b:17:44:8b:0e:ee:a2:bb:
                    0b:03:50:ad:80:9c:05:00:c1:68:69:2f:ac:b9:c9:
                    82:13:fd:22:67
                ASN1 OID: prime256v1
        X509v3 extensions:
            X509v3 Key Usage:
                Digital Signature, Certificate Sign, CRL Sign
            X509v3 Basic Constraints: critical
                CA:TRUE
            X509v3 Subject Key Identifier:
                27:AC:2F:C5:70:00:96:B7:85:E0:6A:BC:9D:E3:BB:EA:8F:8F:05:AA
            1.3.6.1.4.1.311.21.1:
                ...
    Signature Algorithm: ecdsa-with-SHA256
         30:46:02:21:00:d5:ff:e7:89:91:35:83:41:9c:11:29:03:e9:
         49:6f:b4:a8:31:90:a6:b5:8b:4c:1b:87:0f:90:54:3e:f3:69:
         8d:02:21:00:cb:7b:53:d9:58:39:24:aa:d3:33:9b:ce:8f:1e:
         3b:76:d8:2f:e8:92:06:9a:0c:85:d6:0a:90:4e:5a:4f:53:db
```

You can see this one expires at Aug 14 21:18:42 2040 GMT.

If the newly issued server certificate has a new root certificate, then there is a major piece of work to identify every client which connects to the database, and ensure they add the new root certificate in advance to their trusted certificate store. The certificate store (e.g. Oracle Wallet or JKS wallet) can contain multiple trusted certificates, so it is possible to have the old and new root certificates in the store at the same time.

Changing the Server Wallet Password

If the client and server wallet passwords are the same, then need to ensure the server password is changed. While the client wallet password can remain as something easily remembered (as it only contains the trusted root certificate which is publically available within Dell), the server wallet contains the private key used to encrypt all network traffic. For this reason it is vitally important to change this password and secure it in EPV for production.

The wallet password can be changed as follows (assuming the wallet is in /u01/app/oracle/admin/tls_wallet):

```
$ orapki wallet change_pwd -wallet /u01/app/oracle/admin/tls_wallet
Oracle PKI Tool : Version 12.1.0.2
Copyright (c) 2004, 2014, Oracle and/or its affiliates. All rights reserved.

Enter wallet password:
New password:
Enter wallet password:
```

You will be prompted to enter the existing password, followed by the new password.

On production, ensure this password is onboarded to EPV, using the instructions in the GDMS TDE encryption guide here.

Patching Databases using GoldenGate ER or JDBC

Due to the way GoldenGate ER extracts data from a source database, it might hit the following error:

```
OGG-02077  Extract encountered a read error in the asynchronous reader thread and is
abending: Error code 12592, error message: ORA-12592: TNS:bad packet.
```

The database alert log could show the following error at the same time:

```
Wed Jul 05 04:22:32 2017
Exception [type: SIGSEGV, Address not mapped to object] [ADDR:0x7FFD757C6000]
[PC:0x7FEB43042303, __memmove_ssse3_back()+2883] [flags: 0x0, count: 1]
Errors in file /u01/app/oracle/diag/rdbms/ssde/ssde2/trace/ssde2_ora_47790.trc
(incident=176617):
ORA-07445: exception encountered: core dump [__memmove_ssse3_back()+2883] [SIGSEGV]
[ADDR:0x7FFD757C6000] [PC:0x7FEB43042303] [Address not mapped to object] []
ORA-12161: TNS:internal error: partial data received
```

This is exposing a known Oracle bug in 12.1.0.2, which is described in the following Metalink note:

ORA-07445 [memcpy] Reported In 12c For Client Using SSL (Doc ID 2263229.1)

The same bug might also impact JDBC clients using BLOBS:

Application using JDBC/thin with SSL gets "SSL peer shut down incorrectly" when Retrieving Large BLOB (Doc ID 2163050.1)

Both issues are fixed by patch 20544797, so that should be downloaded and applied to the GRID_HOME and ORACLE_HOME.

If there are still issues, then also download and install patch 18841764 described in the following Metalink note:

Bug 18841764 - Network related error like ORA-12592 or ORA-3137 or ORA-3106 may be signaled (Doc ID 18841764.8)

EM13c Monitoring of TLS Enabled Databases

EM13c can monitor listeners and databases that have been configured for TLS. Earlier versions do not support TLS enabled listeners.

When attempting to onboard a TLS enabled database using EM13c, it might report the following error:

```
Server or client wallet has not set properly or Connection Protocol is wrong
```

This could be down to the wallet used by the EM13c OMS server, or client Agent not containing the Dell root certificate used by the database.

The Enterprise Manager Cloud Secure Guide here, describes the emctl commands which need to be run on both the OMS servers, and Agent to specify the location of the trusted keystore.

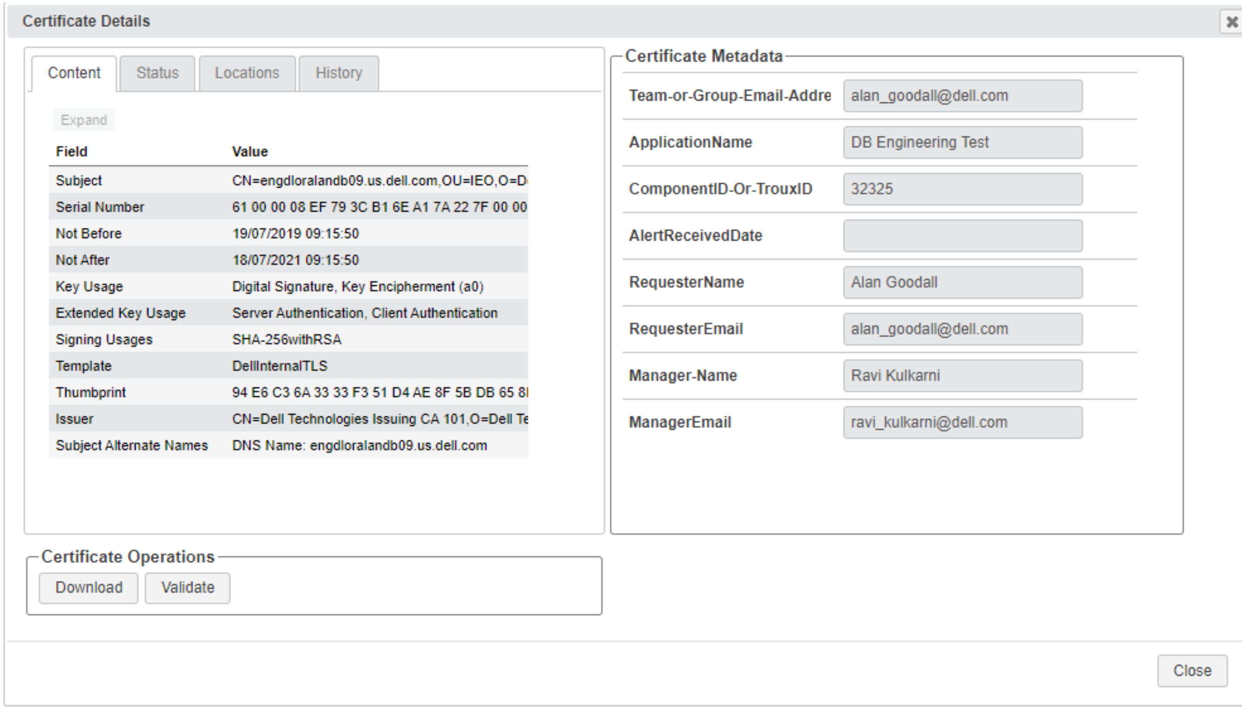
This keystore is a Java keystore, so the existing Java keystore under the OMS and Agent installs needs to be updated, and the Dell Root certificate added as a trusted certificate. The steps are very similar to those described under the Weblogic configuration above.

In both cases (OMS and Agent) it should just be the Server Authority settings that need to reference the keystore, as we are not using client certificates or client based authentication.

Certificate Expiry Dates and Email Contact Details

The Dell issued certificates last for 2 years, and the Dell certificate authority will email the registered email addresses in advance when certificates are due to expire. You can review your certificates in the portal here.

For each certificate, view the certificate details and ensure all the contact information is correctly populated e.g.



Ensure a team or group email address is used, and not personal email address. If any of the details need changed, then email Somika Nayak (or Crypto_Operations if you do not get a timely response).

Appendix A – Useful Certificate/Wallet Commands

This appendix contains useful commands to help query certificates and wallets (assuming you wish to do these steps manually, rather than use the dell_orapki_auto.sh tool). These commands assume there is an environment variable $WALLET pointing to the wallet directory.

To create a new Oracle Wallet (in sub-directory wallet):

```
orapki wallet create -wallet $WALLET -auto_login -pwd welcome1
```

To create a certificate signing request (CSR) in an existing wallet:

```
orapki wallet add -wallet $WALLET -asym_alg RSA -keysize 2048 -dn
"CN=<servername>,OU=<OU name>,O=<ORG name>" -pwd choosesomepassword
```

To export a CSR from a wallet:

```
orapki wallet export -wallet $WALLET -dn "CN=<servername>,OU=<OU name>,O=<ORG name>"
-pwd choosesomepassword -request $WALLET/<servername>.csr –jsafe
```

To display the contents of a CSR:

```
$ openssl req -in <csrfilename.csr> –text -noout
```

To query an Oracle wallet:

```
orapki wallet display -wallet $WALLET -pwd choosesomepassword -jsafe
```

Add a trusted certificate to an Oracle wallet (see the steps below to split a p7b file into the trusted root and intermediate certificates, which can then be imported using this command):

```
orapki wallet add -wallet $WALLET –cert <certfilename> -trusted_cert -pwd
choosesomepassword -jsafe
```

Note that you need to add the trusted in order from top to bottom, i.e. the root certificate, then the intermediate certificate, as the intermediate certificate won't import if the root certificate is not in the wallet.

Add a server certificate to an Oracle wallet which contains the CSR (see the steps below to split a p7b file into the user/server certificates, which can then be imported using this command):

```
orapki wallet add -wallet $WALLET –cert <certfilename> -user_cert -pwd
choosesomepassword -jsafe
```

Ensure the root and intermediate certificates have already been added.

Note that for a client wallet, the user certificate is not required. Only the root certificate is.

To display the contents of a certificate to check its expiry dates and distinguished name (Subject):

```
$ openssl x509 -in   DellRootCertificateAuthority2015.pem -noout -text
```

Split a binary format p7b certificate file into root, intermediate and user certificates, and output in PEM format:

```
openssl pkcs7 -inform DER -in certpkcs7.p7b -print_certs | awk
'split_after==1{n++;split_after=0} /-----END CERTIFICATE-----/ {split_after=1}
{print > "cert" n ".pem"}'
```

This should result in 3 files:

1. cert1.pem – The root certificate
2. cert2.pem – The intermediate certificate
3. cert3.pem – The user/server certificate

To list the contents of a Java JKS keystore:

```
$ORACLE_HOME/jdk/jre/bin/keytool -list -keystore ./cacerts -v -storepass changeit
```

To add a trusted certificate to a Java JKS keystore:

```
$ORACLE_HOME/jdk/jre/bin/keytool -keystore ./cacerts -storepass changeit -importcert
-trustcacerts -alias DellRootCA2015 -file DellRootCertificateAuthority2015.pem
```

Appendix B – Listener Tracing

The database listeners can be traced dynamically by following the instructions in this metalink note:

How to set Oracle Net Listener Tracing Dynamically or at the LSNRCTL Prompt (Doc ID 1940914.1)

Listener tracing can help with troubleshooting blocked listener registration, or connectivity issues. As an example, start lsnrctl, check the location of the trace file, and enable tracing:

```
LSNRCTL> set current_listener LISTENER
Current Listener is LISTENER
LSNRCTL> show trc_file
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=delltest.com)(PORT=1523)))
LISTENER parameter "trc_file" set to ora_17351_47472935529200.trc
The command completed successfully
```

Now enable tracing:

```
LSNRCTL> set trc_level 16
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=delltest.com)(PORT=1523)))
LISTENER parameter "trc_level" set to support
The command completed successfully
```

Now in a separate Putty session, try to connect to the database using SQLPlus, or do a TNSPING using the TNS alias for port 1521.

If they return an ORA- or TNS- error, then review the listener trace file, and search for references to wallets, or ' ntz' (which prefixes many of the SSL related trace output). e.g. the following trace shows an example where the wallet cannot be found because the listener.ora file was missing the WALLET_LOCATION parameter:

```
2017-12-11 09:51:36.099695 : ntzgsvp:no SSL version specified - using default version
0
2017-12-11 09:51:36.099710 : ntzgsvp:exit
2017-12-11 09:51:36.099729 : ntzgcpp:entry
2017-12-11 09:51:36.099748 : ntzgcpp:no SSL cipher suites specified
2017-12-11 09:51:36.099762 : ntzgcpp:exit
2017-12-11 09:51:36.099777 : ntzgcap:entry
2017-12-11 09:51:36.099795 : ntzgcap:no value for client authentication parameter
specified -           using default value: "TRUE"
2017-12-11 09:51:36.099810 : ntzgcap:exit
2017-12-11 09:51:36.099825 : ntzConfigure:client authentication is required.
2017-12-11 09:51:36.099841 : ntzgwrl:entry
2017-12-11 09:51:36.099857 : ntzgwrlFromFile:entry
2017-12-11 09:51:36.099871 : ntzGetStringParameter:entry
2017-12-11 09:51:36.099887 : ntzGetStringParameter:exit
2017-12-11 09:51:36.099902 : ntzGetStringParameter:entry
2017-12-11 09:51:36.099917 : ntzGetStringParameter:exit
2017-12-11 09:51:36.099933 : ntzgwrlFromFile:exit
2017-12-11 09:51:36.099948 : ntzgwrl:no value specified for wallet resource locator -
using default value "NULL"
2017-12-11 09:51:36.099963 : ntzgwrl:exit
2017-12-11 09:51:36.099978 : ntzscr:entry
2017-12-11 09:51:36.100019 : ntzGetStringParameter:entry
2017-12-11 09:51:36.100038 : ntzGetStringParameter:exit
2017-12-11 09:51:36.100053 : ntzGetStringParameter:entry
2017-12-11 09:51:36.100069 : ntzGetStringParameter:exit
2017-12-11 09:51:36.100096 : ntzGetStringParameter:entry
```

```
2017-12-11 09:51:36.100134 : ntzGetStringParameter:exit
2017-12-11 09:51:36.100154 : ntzscr:exit
2017-12-11 09:51:36.100185 : ntzlogin:entry
2017-12-11 09:51:36.100816 : ntzlogin:Wallet open failed with error 28759
2017-12-11 09:51:36.100861 : ntzlogin:returning NZ error 28759 in result structure
2017-12-11 09:51:36.100880 : ntzlogin:failed with error 540
2017-12-11 09:51:36.100896 : ntzlogin:exit
2017-12-11 09:51:36.100911 : ntzConfigure:failed with error 540
2017-12-11 09:51:36.100928 : ntzConfigure:exit
2017-12-11 09:51:36.100944 : ntzCreateConnection:failed with error 540
2017-12-11 09:51:36.100959 : ntzCreateConnection:exit
```

Tracing can be disabled by setting the level back to 0:

```
LSNRCTL> set trc_level 0
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=delltest.com)(PORT=1523)))
LISTENER parameter "trc_level" set to off
The command completed successfully
```

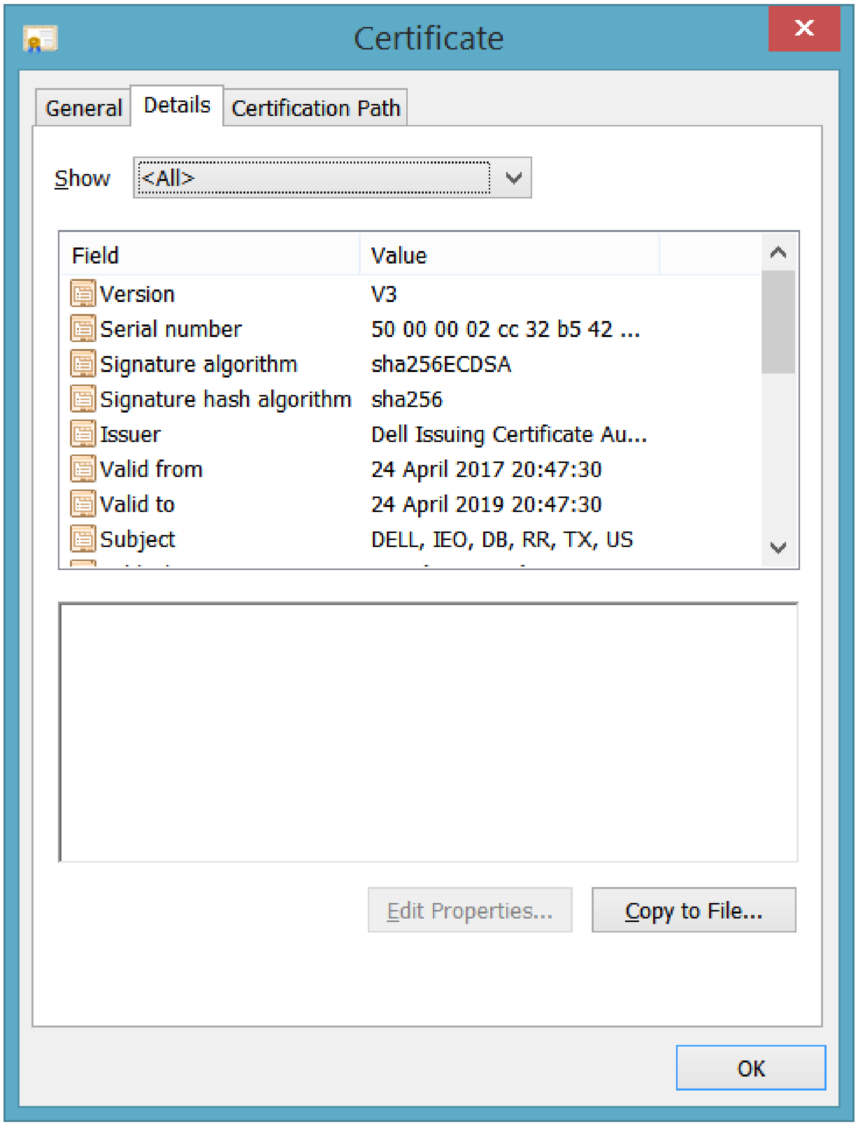
Further SSL troubleshooting tips can be found in this Metalink Note:

SSL Troubleshooting Guide (Doc ID 166492.1)

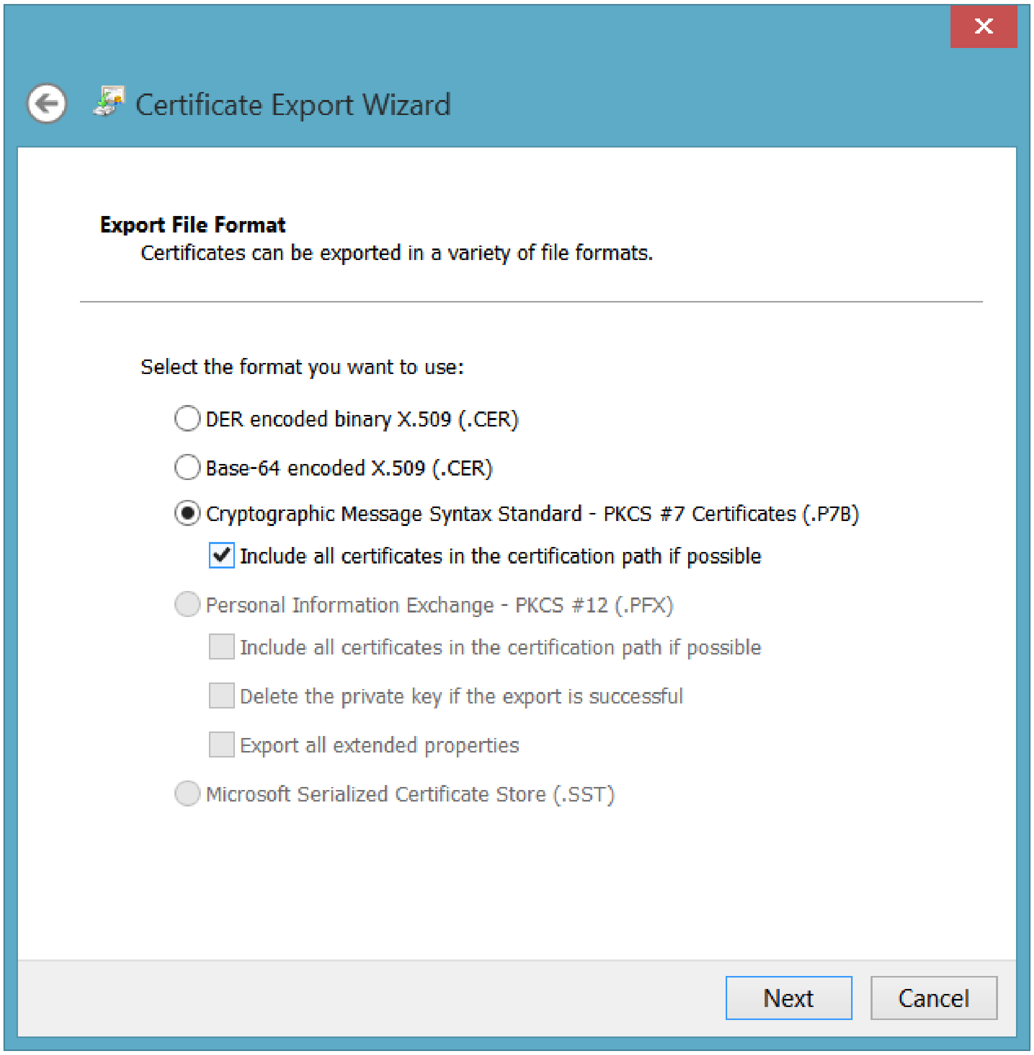
Appendix C – Creating p7b Certificate File using Windows Certificate Manager

If the certificate is not in a p7b format, then the Windows Certificate view can be used to convert it.

Open the .cer file by double clicking on it and navigate to the 'Details' tab and click on the 'Coy to File...' button:



This will bring up the 'Certificate Export Wizard'. Click 'Next' and then select the 'Cryptographic Message Syntax Standard – PKCS #7 Certificates (.P7B)' option. Make sure you check the box 'Include all certificates in the certification path if possible' box:



Click next and save the file with a .p7b extension. Choose a recognisable filename like *custername*.p7b. There will be a confirmation window, so click 'Finish' to complete the export.

Appendix D – Error Codes

The following table shows the typical error codes and how to investigate/fix them.

Error Code	Description/Fix
ORA-29003: SSL transport detected mismatched server certificate	This is usually caused by the DN in the connection string not matching the DN in the server wallet. Check the following: • Ensure sqlnet.ora file is present in the TNS_ADMIN location ($TNS_ADMIN on Linux or %TNS_ADMIN% on Windows), and that it has an entry WALLET_LOCATION. • Check that the parameter in the sqlnet.ora file SSL_SERVER_DN_MATCH is set to TRUE. This forces the DN in the connection string to match the DN in the server wallet. A simple workaround to this error is to set this parameter to FALSE, but that isn't recommended as it wouldn't validate that you are connecting to the correct server. • Check the DN in the server wallet by running the orapki wallet display command. • Check that the DN in the TNS connect string (in the "SECURITY=(SSL_SERVER_CERT_DN=" part of the connect string) matches the DN in the server wallet. See Oracle Support doc id 166492.1.
ORA-29024:Certificate Validation Failure	This is usually an error with the client configuration, so check the following: • Ensure sqlnet.ora file is present in the TNS_ADMIN location ($TNS_ADMIN on Linux or %TNS_ADMIN% on Windows), and that it has an entry WALLET_LOCATION. • Run the c against the wallet in WALLET_LOCATION, and ensure it has a trusted root certificate which matches the root certificate on the server. See Oracle Support doc id 756978.1.
ORA-12520: TNS:listener could not find available handler for requested type of server	This often happens after the TCPS configuration has been applied to a server or cluster. Review all the steps have been completed successfully, and then check the following: • Run a 'lsnrctl services <scanlistener> command against the scan listener (or listener on an Oracle Restart environment). • Look for the entry for the database service and see if the state is 'blocked'. If all the configuration has been done correctly, then this indicates that the database will need restarted to pick up the upgraded sqlnet.ora configuration.
ORA-28860: Fatal SSL error	This can be caused if a 12.1 Oracle client is being used. That needs patch 18604692 applied so that it has support for the Dell approved ciphers. The workaround is to comment the SSL_CIPHER_SUITES parameter in the listener.ora and sqlnet.ora. That breaks the Dell

	cipher security standards though, so either patch the Oracle client (January 2019 PSU or later) or upgrade the client to 19c. This can also be caused by the client trying to use the certificate to authenticate. We only use the certificate to encrypt the SQLNet traffic with TLS, and not to authenticate. Check the following: • Ensure the parameter SSL_CLIENT_AUTHENTICATION is set to FALSE in the client and servers sqlnet.ora. See Oracle Support doc id 843500.1. SSL/TCPS Connection Is Failing With ORA-28860 (Doc ID 843500.1)
ORA-28862: SSL connection failed	This is usually an indication of a firewall issue. If there is an openssl client available, then there is a quick test to see if the firewall is blocking the SSL connection negotiation. Substitute the server or scan/cname into the following command: openssl s_client -connect scanorserverfqdn:1523 -tls1_2 CONNECTED(00000003) write:errno=0 --- no peer certificate available --- No client certificate CA names sent --- SSL handshake has read 0 bytes and written 176 bytes Verification: OK --- New, (NONE), Cipher is (NONE) Secure Renegotiation IS NOT supported If it shows the SSL handshake has read 0 bytes, then the firewall is blocking the connection. Contact the DB engineering team for advice on this process. (openssl is available on all Linux servers, and if you install Ubuntu under Windows 10 you can run this from your laptop).
SQL*Plus **…** **Segmentation fault**	If SQL*Plus hits a segmentation fault and the database is in a DMZ (e.g. in the PCI DMZ) then it is likely that the firewall rules haven't been properly configured for TCPS/1523. This is likely to show as these symptoms: 1. openssl to the scan IP's or VIPs works correctly, and the TLS handshake is successful.

	2. sqlplus using a connection string using the scan names (of the primary and secondary cluster) fails with a segmentation fault. 3. sqlplus using a connection string using a VIP connecting to a database service running on the server where the VIP is running is successful. This is an indication that the sqlnet handshake by which Oracle hands off the connection from the scan to the VIP, which the client then connects to is being blocked somewhere by the firewall (FYI this Oracle document shows the connection process when using a scan). Work with the network security team to ensure that all DB servers, DB VIP's and SCAN IP's are included in the firewall rules. Ask the team to check for any dropped packets in the firewall logs between the client and the database servers.

Contact Details

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